Online Appendix

Escaping the Trade War: Finance and Relational Supply Chains in the Adjustment to Trade Policy Shocks

Felipe Benguria and Felipe Saffie

A Appendix

A.1 Appendix to Section 2

A.1.1 Baseline model

Regarding the response of exports to China to changes in China's tariff, we can use the expression for an exporter's revenue in the Chinese market in equation (4) to write:

$$\frac{\text{dlog}r_{e,CHN}}{\text{dlog}\tau_{CHN}} = (1-\sigma)\cdot\frac{\left(A_{CHN}\cdot\tau_{CHN}^{1-\sigma}\left(\frac{\lambda+1}{\lambda\sigma+1}\right)+A_{ROW}\cdot\tau_{ROW}^{1-\sigma}\right)}{\left(A_{CHN}\cdot\tau_{CHN}^{1-\sigma}+A_{ROW}\cdot\tau_{ROW}^{1-\sigma}\right)}<0 \tag{13}$$

given that $\sigma > 1$.

Regarding the response of exports to the rest of the world to changes in China's tariff, we can use the expression for an exporter's revenue in the rest of the world market in equation (5) to write:

$$\frac{\text{dlog}r_{e,ROW}}{\text{dlog}\tau_{CHN}} = \frac{\lambda(\sigma-1)^2}{\lambda\sigma+1}\cdot A_{CHN}\cdot\tau_{CHN}^{1-\sigma}\cdot\frac{1}{\left(A_{CHN}\cdot\tau_{CHN}^{1-\sigma}+A_{ROW}\cdot\tau_{ROW}^{1-\sigma}\right)}>0 \tag{14}$$

This proves propositions 1a and 1b.

A.1.2 Financial constraints

As discussed in the main text, we assumed that in the initial equilibrium (before China's tariff is raised) a US exporter that sells to any importer in China will also sell to all importers in the rest of the world. A sufficient condition for this is a high enough fixed cost of exporting to China, and we provide a mathematical expression for this assumption below. We focus, without loss of generality, in the case of an exporter that sells to all importers in China (and all importers in the rest of the world) in the initial equilibrium.

We proceed in the following order. First, we show that a higher working capital constraint (associated with a higher leverage ratio as we show later) implies that an exporter is more likely to terminate a trading relationship with an importer in response to an increase in China's tariff. Second, we show that exports to China decline in response to an increase in China's tariff, and that this decline is larger in the case a relationship is terminated. Third, we show that

terminating a relationship in China leads to an increase in exports to the rest of the world. Fourth, we show that a higher working capital constraint is associated with a higher leverage ratio.

■ An exporting firm's profits obtained from selling to all importers take the form:[35]

$$\pi_e^{(N_{CHN},N_{ROW})} = k_\pi \cdot \left(A_{CHN} \cdot \tau_{CHN}^{1-\sigma} \sum_{i=1}^{N_{CHN}} \psi_i^{CHN} + A_{ROW} \cdot \tau_{ROW}^{1-\sigma} \sum_{i=1}^{N_{ROW}} \psi_i^{ROW} \right)^{\frac{\lambda+1}{\lambda\sigma+1}} - \left(1+\frac{\delta}{\nu}\right)(N_{CHN} \cdot g_{CHN} + N_{ROW} \cdot g_{ROW} + f_{CHN} + f_{ROW}) + \frac{(1-\nu)}{\nu}\chi \quad (15)$$

Note that the fixed cost is increasing in the working capital requirement δ.

A sufficiently large tariff can lead the exporter to terminate the relationship with the least productive Chinese importer. This decision is made comparing the profits from selling to all importers against the profits excluding the least productive Chinese importer. The profits obtained from selling to all importers except the least productive Chinese importer are:

$$\pi_e^{(N_{CHN}-1,N_{ROW})} = k_\pi \cdot \left(A_{CHN} \cdot \tau_{CHN}^{1-\sigma} \sum_{i=1}^{N_{CHN}-1} \psi_i^{CHN} + A_{ROW} \cdot \tau_{ROW}^{1-\sigma} \sum_{i=1}^{N_{ROW}} \psi_i^{ROW} \right)^{\frac{\lambda+1}{\lambda\sigma+1}} - \left(1+\frac{\delta}{\nu}\right)((N_{CHN}-1) \cdot g_{CHN} + N_{ROW} \cdot g_{ROW} + f_{CHN} + f_{ROW}) + \frac{(1-\nu)}{\nu}\chi \quad (16)$$

In these expressions we have indexed Chinese importers in decreasing order of productivity, such that $i = N_{CHN}$ corresponds to the lowest productivity importer.

Terminating the trading relationship with the least productive Chinese importer will save the exporter the per-relationship fixed cost g_{CHN}, and will make the exporter face a decline in operating profits (revenue minus variable cost).[36] The fixed–cost saved does not depend on

[35] Note that k_π absorbs the exporter's productivity parameter.

[36] The decline in operating profits is the difference between

$$k_\pi \cdot \left(A_{CHN} \cdot \tau_{CHN}^{1-\sigma} \sum_{i=1}^{N_{CHN}-1} \psi_i^{CHN} + A_{ROW} \cdot \tau_{ROW}^{1-\sigma} \sum_{i=1}^{N_{ROW}} \psi_i^{ROW} \right)^{\frac{\lambda+1}{\lambda\sigma+1}}$$

and

$$k_\pi \cdot \left(A_{CHN} \cdot \tau_{CHN}^{1-\sigma} \sum_{i=1}^{N_{CHN}} \psi_i^{CHN} + A_{ROW} \cdot \tau_{ROW}^{1-\sigma} \sum_{i=1}^{N_{ROW}} \psi_i^{ROW} \right)^{\frac{\lambda+1}{\lambda\sigma+1}}$$

.

the Chinese tariff level. In contrast, the decline in operating profits is smaller under a higher Chinese tariff. Consequently, a higher tariff can lead to terminating a relationship. In other words, the exporter will choose to terminate its relationship with the least productive Chinese importer if this change in profits is positive:

$$\pi_e^{(N_{CHN}-1,N_{ROW})}(\tilde{\tau}_{CHN}) - \pi_e^{(N_{CHN},N_{ROW})}(\tau_{CHN}) > 0\,, \tag{17}$$

where τ_{CHN} and $\tilde{\tau}_{CHN}$ (with a tilde) denote the initial (low) level and the trade war (high) level of the Chinese tariff respectively.

A higher working capital requirement leads to a larger fixed cost (due to the financial component of the fixed cost). On the other hand, the decline in operating profits from terminating a relationship due to the increase in China's tariff is independent of the working capital requirement. Consequently, an exporter is more likely to terminate a relationship with a Chinese importer when the fixed cost is higher (i.e. under a higher working capital constraint).

■ Now, we show that a higher Chinese tariff indeed lowers exports to China if a relationship is terminated.

First, note that for an exporter that sells to all importers in China and the rest of the world, exports to one importer in China are:

$$r_{ei,CHN}^{(N_{CHN},N_{ROW})} = k_r \cdot A_{CHN} \cdot \psi_i^{CHN} \cdot \tau_{CHN}^{1-\sigma} \cdot \left(A_{CHN} \cdot \tau_{CHN}^{1-\sigma} \sum_{i=1}^{N_{CHN}} \psi_i^{CHN} + A_{ROW} \cdot \tau_{ROW}^{1-\sigma} \sum_{i=1}^{N_{ROW}} \psi_i^{ROW} \right)^{\frac{-\lambda(\sigma-1)}{\lambda\sigma+1}} \tag{18}$$

and exports to all importers in China are:

$$r_{e,CHN}^{(N_{CHN},N_{ROW})} = k_r \left(A_{CHN} \tau_{CHN}^{1-\sigma} \sum_{i=1}^{N_{CHN}} \psi_i^{CHN} \right) \cdot \left(A_{CHN} \cdot \tau_{CHN}^{1-\sigma} \sum_{i=1}^{N_{CHN}} \psi_i^{CHN} + A_{ROW} \cdot \tau_{ROW}^{1-\sigma} \sum_{i=1}^{N_{ROW}} \psi_i^{ROW} \right)^{\frac{-\lambda(\sigma-1)}{\lambda\sigma+1}} \tag{19}$$

In these expressions, $k_r = \left(\frac{\sigma-1}{\sigma}\right)^{\frac{\sigma-1}{\lambda\sigma+1}} \cdot \varphi_e^{\frac{\sigma-1}{\lambda\sigma+1}}$.

Terminating the relationship with the least productive importer leads to a decline in exports

to China if:

$$k_r \cdot \left(A_{CHN} \tilde{\tau}_{CHN}^{1-\sigma} \sum_{i=1}^{N_{CHN}-1} \psi_i^{CHN} \right) \cdot \left(A_{CHN} \cdot \tilde{\tau}_{CHN}^{1-\sigma} \sum_{i=1}^{N_{CHN}-1} \psi_i^{CHN} + A_{ROW} \cdot \tau_{ROW}^{1-\sigma} \sum_{i=1}^{N_{ROW}} \psi_i^{ROW} \right)^{-\frac{\lambda(\sigma-1)}{\lambda\sigma+1}}$$
$$< k_r \cdot \left(A_{CHN} \tau_{CHN}^{1-\sigma} \sum_{i=1}^{N_{CHN}} \psi_i^{CHN} \right) \cdot \left(A_{CHN} \cdot \tau_{CHN}^{1-\sigma} \sum_{i=1}^{N_{CHN}} \psi_i^{CHN} + A_{ROW} \cdot \tau_{ROW}^{1-\sigma} \sum_{i=1}^{N_{ROW}} \psi_i^{ROW} \right)^{-\frac{\lambda(\sigma-1)}{\lambda\sigma+1}} \tag{20}$$

This can be rearranged as:

$$\frac{\left(A_{CHN} \tilde{\tau}_{CHN}^{1-\sigma} \sum_{i=1}^{N_{CHN}-1} \psi_i^{CHN} \right)}{\left(A_{CHN} \tau_{CHN}^{1-\sigma} \sum_{i=1}^{N_{CHN}} \psi_i^{CHN} \right)} < \frac{\left(A_{CHN} \cdot \tilde{\tau}_{CHN}^{1-\sigma} \sum_{i=1}^{N_{CHN}-1} \psi_i^{CHN} + A_{ROW} \cdot \tau_{ROW}^{1-\sigma} \sum_{i=1}^{N_{ROW}} \psi_i^{ROW} \right)^{\frac{\lambda(\sigma-1)}{\lambda\sigma+1}}}{\left(A_{CHN} \cdot \tau_{CHN}^{1-\sigma} \sum_{i=1}^{N_{CHN}} \psi_i^{CHN} + A_{ROW} \cdot \tau_{ROW}^{1-\sigma} \sum_{i=1}^{N_{ROW}} \psi_i^{ROW} \right)^{\frac{\lambda(\sigma-1)}{\lambda\sigma+1}}} \tag{21}$$

The inequality holds because the left hand side is smaller than one and the right hand side i) adds the same positive term to the numerator and denominator and ii) has an exponent that is positive and lower than one (each of these reasons is sufficient for the right hand side to be larger).

■ The decline in exports to China is larger if a relationship is terminated than if it is not. If a relationship is terminated, exports to China fall from the expression in (19) to:

$$r_{e,CHN}^{(N_{CHN}-1,N_{ROW})} = k_r \left(A_{CHN} \tilde{\tau}_{CHN}^{1-\sigma} \sum_{i=1}^{N_{CHN}-1} \psi_i^{CHN} \right) \cdot \left(A_{CHN} \cdot \tilde{\tau}_{CHN}^{1-\sigma} \sum_{i=1}^{N_{CHN}-1} \psi_i^{CHN} + A_{ROW} \cdot \tau_{ROW}^{1-\sigma} \sum_{i=1}^{N_{ROW}} \psi_i^{ROW} \right)^{\frac{-\lambda(\sigma-1)}{\lambda\sigma+1}} \tag{22}$$

If a relationship is not terminated, exports to China fall from the expression in (19) to:

$$r_{e,CHN}^{(N_{CHN},N_{ROW})} = k_r \left(A_{CHN} \tilde{\tau}_{CHN}^{1-\sigma} \sum_{i=1}^{N_{CHN}} \psi_i^{CHN} \right) \cdot \left(A_{CHN} \cdot \tilde{\tau}_{CHN}^{1-\sigma} \sum_{i=1}^{N_{CHN}} \psi_i^{CHN} + A_{ROW} \cdot \tau_{ROW}^{1-\sigma} \sum_{i=1}^{N_{ROW}} \psi_i^{ROW} \right)^{\frac{-\lambda(\sigma-1)}{\lambda\sigma+1}} \tag{23}$$

That (22) is lower than (23) implies that:

$$\frac{\sum_{i=1}^{N_{CHN-1}} \psi_i^{CHN}}{\sum_{i=1}^{N_{CHN}} \psi_i^{CHN}} < \frac{\left(A_{CHN}\tilde{\tau}_{CHN}^{1-\sigma}\sum_{i=1}^{N_{CHN}-1}\psi_i^{CHN} + A_{ROW}\cdot\tau_{ROW}^{1-\sigma}\sum_{i=1}^{N_{ROW}}\psi_i^{ROW}\right)^{\frac{\lambda(\sigma-1)}{\lambda\sigma+1}}}{\left(A_{CHN}\tilde{\tau}_{CHN}^{1-\sigma}\sum_{i=1}^{N_{CHN}}\psi_i^{CHN} + A_{ROW}\cdot\tau_{ROW}^{1-\sigma}\sum_{i=1}^{N_{ROW}}\psi_i^{ROW}\right)^{\frac{\lambda(\sigma-1)}{\lambda\sigma+1}}} \tag{24}$$

The inequality holds because the left hand side is smaller than one and the right hand side i) multiplies the numerator and denominator by the same positive term and adds the same positive term to both ii) has an exponent that is positive and lower than one (each of these reasons is sufficient for the right hand side to be larger).

■ Note that it is straightforward to generalize this argument for the case of a larger tariff such that the tariff leads to terminating relationships with any number of Chinese importers.

■ We have argued that a higher working capital requirement, which entails a higher fixed cost, makes terminating relationships in response of an increase in China's tariff more likely. We have showed that this leads to a larger decline in exports to China. For this reason, it will also lead to a larger decline in marginal cost. As shown in the discussion for the baseline model, this leads to a larger increase in exports to the rest of the world.

Exports to the rest of the world for an exporter that sells to all importers in China and the rest of the world are:

$$r_{e,ROW}^{(N_{CHN},N_{ROW})} = k_r\left(A_{ROW}\tau_{ROW}^{1-\sigma}\sum_{i=1}^{N_{ROW}}\psi_i^{ROW}\right)\cdot\left(A_{CHN}\cdot\tau_{CHN}^{1-\sigma}\sum_{i=1}^{N_{CHN}}\psi_i^{CHN} + A_{ROW}\cdot\tau_{ROW}^{1-\sigma}\sum_{i=1}^{N_{ROW}}\psi_i^{ROW}\right)^{\frac{-\lambda(\sigma-1)}{\lambda\sigma+1}} \tag{25}$$

The increase in exports to the rest of the world is larger is larger if a relationship in China is terminated than if it is not. If a relationship in China is terminated as a result of the higher tariffs, exports to the rest of the world increase from the expression in (25) to:

$$\begin{aligned} r_{e,ROW}^{(N_{CHN}-1,N_{ROW})} = {} & k_r\left(A_{ROW}\tau_{ROW}^{1-\sigma}\sum_{i=1}^{N_{ROW}}\psi_i^{ROW}\right) \\ & \cdot\left(A_{CHN}\cdot\tilde{\tau}_{CHN}^{1-\sigma}\sum_{i=1}^{N_{CHN}-1}\psi_i^{CHN} + A_{ROW}\cdot\tau_{ROW}^{1-\sigma}\sum_{i=1}^{N_{ROW}}\psi_i^{ROW}\right)^{\frac{-\lambda(\sigma-1)}{\lambda\sigma+1}} \end{aligned} \tag{26}$$

If a relationship in China is not terminated, exports to the rest of the world increase from the

expression in (25) to:

$$r_{e,ROW}^{(N_{CHN},N_{ROW})} = k_r \left(A_{ROW} \tau_{ROW}^{1-\sigma} \sum_{i=1}^{N_{ROW}} \psi_i^{ROW} \right) \cdot \left(A_{CHN} \cdot \tilde{\tau}_{CHN}^{1-\sigma} \sum_{i=1}^{N_{CHN}} \psi_i^{CHN} + A_{ROW} \cdot \tau_{ROW}^{1-\sigma} \sum_{i=1}^{N_{ROW}} \psi_i^{ROW} \right)^{\frac{-\lambda(\sigma-1)}{\lambda\sigma+1}} \tag{27}$$

We can see from inspection of (26) and (27) that the case in which a relationship in China is terminated leads to a higher increase in exports in the rest of the world (because (27) has an extra positive term inside the second parenthesis, and the second parenthesis has a negative exponent).

■ We have assumed that in the initial equilibrium (before tariffs are raised), if US exporters export to any importer in China, they also export to all importers in the rest of the world. In other words, that China is a harder market to export to. We show below that a sufficient condition for this is that the fixed cost of exporting to China is sufficiently high.

For this to hold, an exporter that already exports to $N_{ROW} - 1$ importers in ROW will prefer (i.e. will receive a larger increase in profits from) to export to the least productive importer in ROW than to the most productive importer in China.

$$\pi_e^{(0,N_{ROW})} - \pi_e^{(0,N_{ROW}-1)} > \pi_e^{(1,N_{ROW}-1)} - \pi_e^{(0,N_{ROW}-1)} \tag{28}$$

This implies:

$$\pi_e^{(0,N_{ROW})} > \pi_e^{(1,N_{ROW}-1)} \tag{29}$$

We can write

$$\pi_e^{(0,N_{ROW})} = k \cdot \left(A_{ROW} \cdot \tau_{ROW}^{1-\sigma} \cdot \sum_{i=1}^{N_{ROW}} \psi_i^{ROW} \right)^{\frac{\lambda+1}{\lambda\sigma+1}} - N_{ROW} \cdot g_{ROW} - f_{ROW} \tag{30}$$

and

$$\begin{aligned} \pi_e^{(1,N_{ROW}-1)} = k \cdot & \left(A_{CHN} \cdot \tau_{CHN}^{1-\sigma} \cdot \psi_1^{CHN} + A_{ROW} \cdot \tau_{ROW}^{1-\sigma} \cdot \sum_{i=1}^{N_{ROW}-1} \psi_i^{ROW} \right)^{\frac{\lambda+1}{\lambda\sigma+1}} \\ & - g_{CHN} - (N_{ROW} - 1) \cdot g_{ROW} - f_{CHN} - f_{ROW} \end{aligned} \tag{31}$$

Upon comparison of the last two expressions, if we assume that initial tariffs are the same in both markets and the per–relationship fixed cost is also the same in both markets, a sufficiently large fixed cost of selling to the Chinese market (f_{CHN}) will be sufficient for this assumption to hold. Specifically, this would be the case if:

$$f_{CHN} > k \cdot \left(A_{CHN} \cdot \tau_{CHN}^{1-\sigma} \cdot \psi_1^{CHN} + A_{ROW} \cdot \tau_{ROW}^{1-\sigma} \cdot \sum_{i=1}^{N_{ROW}-1} \psi_i^{ROW} \right)^{\frac{\lambda+1}{\lambda\sigma+1}} - k \cdot \left(A_{ROW} \cdot \tau_{ROW}^{1-\sigma} \cdot \sum_{i=1}^{N_{ROW}} \psi_i^{ROW} \right)^{\frac{\lambda+1}{\lambda\sigma+1}} \tag{32}$$

■ Finally, note that a higher fraction δ of the total cost to be financed implies a higher leverage ratio. The leverage ratio is defined as the ratio between the amount borrowed and the firm's revenue and in the initial equilibrium it is:

$$L_e = \frac{\delta \cdot \tilde{f}}{r_{e,CHN}^{(N_{CHN},N_{ROW})} + r_{e,ROW}^{(N_{CHN},N_{ROW})}} \tag{33}$$

where $\tilde{f}$ represents the sum of the fixed costs of exporting (including the per–relationship fixed costs but not the financial component resulting from the working capital requirement). A small increase in δ that does not lead to an adjustment along the subextensive margin (adding or terminating relationships) implies an increase in the numerator and no change in the denominator. Now consider a larger increase in δ that leads to terminating a trading relationship. A relationship is terminated when the decline in operating profits (revenue minus variable cost) from doing so is larger than the per–relationship fixed cost the exporter saves on. The decline in revenue is larger than the decline in profits, and consequently, larger than the per–relationship fixed cost the exporter stops paying. Thus, the increase in δ leads to a larger decline in the denominator than in the numerator, and is associated with a higher leverage ratio.

A.1.3 Relationship stickiness

In the case of financial constraints analyzed in the previous subsection, we assumed that in the initial equilibrium (before China's tariff is raised) a US exporter that sells to any importer in China will also sell to all importers in the rest of the world. This assumption is not necessary in

the case of relationship stickiness, but we keep it here for ease of exposition, even though the proofs generalize to any initial equilibrium. We focus, without loss of generality, in the case of an exporter that sells to all importers in China (and all importers in the rest of the world) in the initial equilibrium.

Consider an increase in Chinese tariffs that is large enough such that the exporter's profits are higher if it sells to all Chinese importers except the least productive one. Assume, on the other hand, that the increase in tariffs is not too large such that the US exporter still finds it profitable to sell to all other Chinese importers. (In other words, the exporter sells to the N_{ROW} rest of the world importers and to $N_{CHN} - 1$ Chinese importers).[37]

An exporting firm's profits obtained from selling to all importers take the form:[38]

$$\pi_e^{(N_{CHN},N_{ROW})} = k_\pi \cdot \left(A_{CHN} \cdot \tau_{CHN}^{1-\sigma} \sum_{i=1}^{N_{CHN}} \psi_i^{CHN} + A_{ROW} \cdot \tau_{ROW}^{1-\sigma} \sum_{i=1}^{N_{ROW}} \psi_i^{ROW} \right)^{\frac{\lambda+1}{\lambda\sigma+1}} - N_{CHN} \cdot g_{CHN} - N_{ROW} \cdot g_{ROW} - f_{CHN} - f_{ROW} \tag{34}$$

and the profits obtained from selling to all importers except the least productive Chinese importer are:

$$\pi_e^{(N_{CHN}-1,N_{ROW})} = k_\pi \cdot \left(A_{CHN} \cdot \tau_{CHN}^{1-\sigma} \sum_{i=1}^{N_{CHN}-1} \psi_i^{CHN} + A_{ROW} \cdot \tau_{ROW}^{1-\sigma} \sum_{i=1}^{N_{ROW}} \psi_i^{ROW} \right)^{\frac{\lambda+1}{\lambda\sigma+1}} - (N_{CHN} - 1) \cdot g_{CHN} - N_{ROW} \cdot g_{ROW} - f_{CHN} - f_{ROW} \tag{35}$$

In these expressions we have indexed Chinese importers in decreasing order of productivity, such that $i = N_{CHN}$ corresponds to the lowest productivity importer.

Terminating the trading relationship with the least productive Chinese importer will save the exporter the per-relationship fixed cost g_{CHN}, and will make the exporter face a decline in operating profits (revenue minus variable cost).[39] The fixed–cost saved does not depend on

[37] It is straightforward to generalize the argument for a larger tariff increase such that multiple relationships become unprofitable. If the tariff is smaller than this, relationship stickiness would not play a role.

[38] Note that k_π absorbs the exporter's productivity parameter.

[39] The decline in operating profits is the difference between

$$k_\pi \cdot \left(A_{CHN} \cdot \tau_{CHN}^{1-\sigma} \sum_{i=1}^{N_{CHN}-1} \psi_i^{CHN} + A_{ROW} \cdot \tau_{ROW}^{1-\sigma} \sum_{i=1}^{N_{ROW}} \psi_i^{ROW} \right)^{\frac{\lambda+1}{\lambda\sigma+1}}$$

the Chinese tariff level. In contrast, the decline in operating profits is smaller under a higher Chinese tariff. Consequently, a higher tariff can lead to terminating a relationship.

When there is a cost of terminating a relationship (denoted Γ), the exporter will choose to terminate its relationship with the least productive Chinese importer if this change in profits is larger than the termination cost Γ:

$$\pi_e^{(N_{CHN}-1,N_{ROW})}(\tilde{\tau}_{CHN}) - \pi_e^{(N_{CHN},N_{ROW})}(\tau_{CHN}) > \Gamma\,, \tag{36}$$

where τ_{CHN} and $\tilde{\tau}_{CHN}$ (with a tilde) denote the initial (low) level and the trade war (high) level of the Chinese tariff respectively.

If the termination cost Γ is low (i.e. if relationship stickiness is low), the US exporter will indeed terminate the relationship with the least productive Chinese importer. If the termination cost Γ is high (i.e. if relationship stickiness is high), the US exporter will not terminate the relationship with the least productive Chinese importer.

■ Now, we show that a higher Chinese tariff indeed lowers exports to China if a relationship is terminated.

First, note that for an exporter that sells to all importers in China and the rest of the world, exports to one importer in China are:

$$r_{ei,CHN}^{(N_{CHN},N_{ROW})} = k_r \cdot A_{CHN} \cdot \psi_i^{CHN} \cdot \tau_{CHN}^{1-\sigma} \cdot \left(A_{CHN} \cdot \tau_{CHN}^{1-\sigma} \sum_{i=1}^{N_{CHN}} \psi_i^{CHN} + A_{ROW} \cdot \tau_{ROW}^{1-\sigma} \sum_{i=1}^{N_{ROW}} \psi_i^{ROW} \right)^{\frac{-\lambda(\sigma-1)}{\lambda\sigma+1}} \tag{37}$$

and exports to all importers in China are:

$$r_{e,CHN}^{(N_{CHN},N_{ROW})} = k_r \left(A_{CHN} \tau_{CHN}^{1-\sigma} \sum_{i=1}^{N_{CHN}} \psi_i^{CHN} \right) \cdot \left(A_{CHN} \cdot \tau_{CHN}^{1-\sigma} \sum_{i=1}^{N_{CHN}} \psi_i^{CHN} + A_{ROW} \cdot \tau_{ROW}^{1-\sigma} \sum_{i=1}^{N_{ROW}} \psi_i^{ROW} \right)^{\frac{-\lambda(\sigma-1)}{\lambda\sigma+1}} \tag{38}$$

In these expressions, $k_r = \left(\frac{\sigma-1}{\sigma}\right)^{\frac{\sigma-1}{\lambda\sigma+1}} \cdot \varphi_e^{\frac{\sigma-1}{\lambda\sigma+1}}$.

Terminating the relationship with the least productive importer leads to a decline in exports

and

$$k_\pi \cdot \left(A_{CHN} \cdot \tau_{CHN}^{1-\sigma} \sum_{i=1}^{N_{CHN}} \psi_i^{CHN} + A_{ROW} \cdot \tau_{ROW}^{1-\sigma} \sum_{i=1}^{N_{ROW}} \psi_i^{ROW} \right)^{\frac{\lambda+1}{\lambda\sigma+1}}$$

.

to China if:

$$k_r \cdot \left(A_{CHN} \tilde{\tau}_{CHN}^{1-\sigma} \sum_{i=1}^{N_{CHN}-1} \psi_i^{CHN} \right) \cdot \left(A_{CHN} \cdot \tilde{\tau}_{CHN}^{1-\sigma} \sum_{i=1}^{N_{CHN}-1} \psi_i^{CHN} + A_{ROW} \cdot \tau_{ROW}^{1-\sigma} \sum_{i=1}^{N_{ROW}} \psi_i^{ROW} \right)^{-\frac{\lambda(\sigma-1)}{\lambda\sigma+1}}$$
$$< k_r \cdot \left(A_{CHN} \tau_{CHN}^{1-\sigma} \sum_{i=1}^{N_{CHN}} \psi_i^{CHN} \right) \cdot \left(A_{CHN} \cdot \tau_{CHN}^{1-\sigma} \sum_{i=1}^{N_{CHN}} \psi_i^{CHN} + A_{ROW} \cdot \tau_{ROW}^{1-\sigma} \sum_{i=1}^{N_{ROW}} \psi_i^{ROW} \right)^{-\frac{\lambda(\sigma-1)}{\lambda\sigma+1}} \tag{39}$$

This can be rearranged as:

$$\frac{\left(A_{CHN} \tilde{\tau}_{CHN}^{1-\sigma} \sum_{i=1}^{N_{CHN}-1} \psi_i^{CHN} \right)}{\left(A_{CHN} \tau_{CHN}^{1-\sigma} \sum_{i=1}^{N_{CHN}} \psi_i^{CHN} \right)} < \frac{\left(A_{CHN} \cdot \tilde{\tau}_{CHN}^{1-\sigma} \sum_{i=1}^{N_{CHN}-1} \psi_i^{CHN} + A_{ROW} \cdot \tau_{ROW}^{1-\sigma} \sum_{i=1}^{N_{ROW}} \psi_i^{ROW} \right)^{\frac{\lambda(\sigma-1)}{\lambda\sigma+1}}}{\left(A_{CHN} \cdot \tau_{CHN}^{1-\sigma} \sum_{i=1}^{N_{CHN}} \psi_i^{CHN} + A_{ROW} \cdot \tau_{ROW}^{1-\sigma} \sum_{i=1}^{N_{ROW}} \psi_i^{ROW} \right)^{\frac{\lambda(\sigma-1)}{\lambda\sigma+1}}} \tag{40}$$

The inequality holds because the left hand side is smaller than one and the right hand side i) adds the same positive term to the numerator and denominator and ii) has an exponent that is positive and lower than one (each of these reasons is sufficient for the right hand side to be larger).

■ The decline in exports to China is larger if a relationship is terminated than if it is not. If a relationship is terminated, exports to China fall from the expression in (38) to:

$$r_{e,CHN}^{(N_{CHN}-1,N_{ROW})} = k_r \left(A_{CHN} \tilde{\tau}_{CHN}^{1-\sigma} \sum_{i=1}^{N_{CHN}-1} \psi_i^{CHN} \right) \cdot \left(A_{CHN} \cdot \tilde{\tau}_{CHN}^{1-\sigma} \sum_{i=1}^{N_{CHN}-1} \psi_i^{CHN} + A_{ROW} \cdot \tau_{ROW}^{1-\sigma} \sum_{i=1}^{N_{ROW}} \psi_i^{ROW} \right)^{\frac{-\lambda(\sigma-1)}{\lambda\sigma+1}} \tag{41}$$

If a relationship is not terminated, exports to China fall from the expression in (38) to:

$$r_{e,CHN}^{(N_{CHN},N_{ROW})} = k_r \left(A_{CHN} \tilde{\tau}_{CHN}^{1-\sigma} \sum_{i=1}^{N_{CHN}} \psi_i^{CHN} \right) \cdot \left(A_{CHN} \cdot \tilde{\tau}_{CHN}^{1-\sigma} \sum_{i=1}^{N_{CHN}} \psi_i^{CHN} + A_{ROW} \cdot \tau_{ROW}^{1-\sigma} \sum_{i=1}^{N_{ROW}} \psi_i^{ROW} \right)^{\frac{-\lambda(\sigma-1)}{\lambda\sigma+1}} \tag{42}$$

That (41) is lower than (42) implies that:

$$\frac{\sum_{i=1}^{N_{CHN-1}} \psi_i^{CHN}}{\sum_{i=1}^{N_{CHN}} \psi_i^{CHN}} < \frac{\left(A_{CHN}\tilde{\tau}_{CHN}^{1-\sigma}\sum_{i=1}^{N_{CHN}-1}\psi_i^{CHN} + A_{ROW}\cdot\tau_{ROW}^{1-\sigma}\sum_{i=1}^{N_{ROW}}\psi_i^{ROW}\right)^{\frac{\lambda(\sigma-1)}{\lambda\sigma+1}}}{\left(A_{CHN}\tilde{\tau}_{CHN}^{1-\sigma}\sum_{i=1}^{N_{CHN}}\psi_i^{CHN} + A_{ROW}\cdot\tau_{ROW}^{1-\sigma}\sum_{i=1}^{N_{ROW}}\psi_i^{ROW}\right)^{\frac{\lambda(\sigma-1)}{\lambda\sigma+1}}} \tag{43}$$

The inequality holds because the left hand side is smaller than one and the right hand side i) multiplies the numerator and denominator by the same positive term and adds the same positive term to both ii) has an exponent that is positive and lower than one (each of these reasons is sufficient for the right hand side to be larger).

■ Note that it is straightforward to generalize this argument for the case of a larger tariff such that the tariff leads to terminating relationships with any number of Chinese importers.

■ Because the case of higher relationship stickiness, which can prevent terminating relationships, leads to a smaller decline in exports to China, it will also lead to a smaller decline in marginal cost. As shown in the discussion for the baseline model, this leads to a lower increase in exports to the rest of the world.

Exports to the rest of the world for an exporter that sells to all importers in China and the rest of the world are:

$$r_{e,ROW}^{(N_{CHN},N_{ROW})} = k_r\left(A_{ROW}\tau_{ROW}^{1-\sigma}\sum_{i=1}^{N_{ROW}}\psi_i^{ROW}\right)\cdot\left(A_{CHN}\cdot\tau_{CHN}^{1-\sigma}\sum_{i=1}^{N_{CHN}}\psi_i^{CHN} + A_{ROW}\cdot\tau_{ROW}^{1-\sigma}\sum_{i=1}^{N_{ROW}}\psi_i^{ROW}\right)^{\frac{-\lambda(\sigma-1)}{\lambda\sigma+1}} \tag{44}$$

The increase in exports to the rest of the world is larger if a relationship in China is terminated than if it is not. If a relationship in China is terminated as a result of the higher tariffs, exports to the rest of the world increase from the expression in (44) to:

$$\begin{aligned} r_{e,ROW}^{(N_{CHN}-1,N_{ROW})} = {} & k_r\left(A_{ROW}\tau_{ROW}^{1-\sigma}\sum_{i=1}^{N_{ROW}}\psi_i^{ROW}\right) \\ & \cdot\left(A_{CHN}\cdot\tilde{\tau}_{CHN}^{1-\sigma}\sum_{i=1}^{N_{CHN}-1}\psi_i^{CHN} + A_{ROW}\cdot\tau_{ROW}^{1-\sigma}\sum_{i=1}^{N_{ROW}}\psi_i^{ROW}\right)^{\frac{-\lambda(\sigma-1)}{\lambda\sigma+1}} \end{aligned} \tag{45}$$

If a relationship in China is not terminated, exports to the rest of the world increase from the

expression in (44) to:

$$r_{e,ROW}^{(N_{CHN},N_{ROW})} = k_r \left(A_{ROW} \tau_{ROW}^{1-\sigma} \sum_{i=1}^{N_{ROW}} \psi_i^{ROW} \right) \cdot \left(A_{CHN} \cdot \tilde{\tau}_{CHN}^{1-\sigma} \sum_{i=1}^{N_{CHN}} \psi_i^{CHN} + A_{ROW} \cdot \tau_{ROW}^{1-\sigma} \sum_{i=1}^{N_{ROW}} \psi_i^{ROW} \right)^{\frac{-\lambda(\sigma-1)}{\lambda\sigma+1}} \tag{46}$$

We can see from inspection of (45) and (46) that the case in which a relationship in China is terminated leads to a higher increase in exports in the rest of the world (because (46) has an extra positive term inside the second parenthesis, and the second parenthesis has a negative exponent).

A.1.4 Product Differentiation (elasticity of substitution)

■ Recall from the baseline model that:

$$\frac{dlog(r_{CHN})}{dlog(\tau_{CHN})} = (1-\sigma) \frac{\left(A\tau_A^{1-\sigma} \cdot \left(\frac{1+\lambda}{\lambda\sigma+1} \right) + B \cdot \tau_B^{1-\sigma} \right)}{\left(A\tau_A^{1-\sigma} + B \cdot \tau_B^{1-\sigma} \right)} \tag{47}$$

Define:

$$G = \frac{\left(A\tau_A^{1-\sigma} \cdot \left(\frac{1+\lambda}{\lambda\sigma+1} \right) + B \cdot \tau_B^{1-\sigma} \right)}{\left(A\tau_A^{1-\sigma} + B \cdot \tau_B^{1-\sigma} \right)} \tag{48}$$

such that:

$$\frac{d^2 log(r_{CHN})}{(dlog(\tau_{CHN}))(d\sigma)} = -G + (1-\sigma)\frac{dG}{d\sigma} \tag{49}$$

Call G_{NUM} the numerator of G and G_{DEN} the denominator of G, such that:

$$\frac{dG}{d\sigma} = \frac{\frac{dG_{NUM}}{d\sigma} \cdot G_{DEN} - \frac{dG_{DEN}}{d\sigma} \cdot G_{NUM}}{G_{DEN}^2} \tag{50}$$

We find:

$$\frac{dG_{NUM}}{d\sigma} = -A\tau_A^{1-\sigma} \cdot log(\tau_A) \left(\frac{1+\lambda}{\lambda\sigma+1} \right) - A\tau_A^{1-\sigma} \frac{\lambda(1+\lambda)}{(\lambda\sigma+1)^2} - B\tau_B^{1-\sigma} log(\tau_B) \tag{51}$$

and

$$\frac{dG_{DEN}}{d\sigma} = -A\tau_A^{1-\sigma} \cdot log(\tau_A) - B\tau_B^{1-\sigma} log(\tau_B) \tag{52}$$

Considering that $\sigma > 1$ and simplifying these expressions, we obtain $\frac{d^2 log(r_{CHN})}{(dlog(\tau_{CHN}))(d\sigma)} < 0$, which corresponds to proposition 4a.

■ Recall from the baseline model that:

$$\frac{\text{dlog}r_{e,ROW}}{\text{dlog}\tau_{CHN}} = \frac{\lambda(\sigma-1)^2}{\lambda\sigma+1} \cdot A_{CHN} \cdot \tau_{CHN}^{1-\sigma} \cdot \frac{1}{\left(A_{CHN} \cdot \tau_{CHN}^{1-\sigma} + A_{ROW} \cdot \tau_{ROW}^{1-\sigma}\right)} \tag{53}$$

Define:

$$F = \frac{\lambda(\sigma-1)^2}{\lambda\sigma+1} \tag{54}$$

and

$$G = \frac{A\tau_A^{1-\sigma}}{\left(A\tau_A^{1-\sigma} + B \cdot \tau_B^{1-\sigma}\right)} \tag{55}$$

such that:

$$\frac{d^2 log(r_{CHN})}{(dlog(\tau_{CHN}))(d\sigma)} = \frac{dF}{d\sigma} \cdot G + \frac{dG}{d\sigma} \cdot F \tag{56}$$

We find:

$$\frac{dF}{d\sigma} = \frac{2\lambda(\sigma-1)(\lambda\sigma+1) - \lambda^2(\sigma-1)^2}{(\lambda\sigma+1)^2} \tag{57}$$

Call G_{NUM} the numerator of G and G_{DEN} the denominator of G, such that:

$$\frac{dG}{d\sigma} = \frac{\frac{dG_{NUM}}{d\sigma} \cdot G_{DEN} - \frac{dG_{DEN}}{d\sigma} \cdot G_{NUM}}{G_{DEN}^2} \tag{58}$$

We find:

$$\frac{dG_{NUM}}{d\sigma} = -A\tau_A^{1-\sigma} \cdot log(\tau_A) \tag{59}$$

and

$$\frac{dG_{DEN}}{d\sigma} = -A\tau_A^{1-\sigma} \cdot log(\tau_A) - B\tau_B^{1-\sigma} log(\tau_B) \tag{60}$$

Considering that $\sigma > 1$ and simplifying these expressions, we obtain $\frac{d^2 log(r_{ROW})}{(dlog(\tau_{CHN}))(d\sigma)} > 0$, which corresponds to proposition 4b.

A.1.5 Additional results: Baseline model with ad–valorem tariffs

The standard approach to modeling tariffs in the literature is using the iceberg cost assumption [Melitz, 2003]. As a robustness check, we show are results when modeling tariffs as ad-valorem, departing from the iceberg cost assumption. In the standard version of Melitz [2003] with a constant marginal cost, modeling tariffs as ad–valorem tariffs or iceberg trade costs yields

the same result for prices, quantities and revenue [Cole, 2011]. In our case, with an increasing marginal cost, the results from both approaches differ to some extent, although we show below that the propositions of our baseline model also hold under the ad-valorem modeling approach.

In this case, we write preferences as follows,

$$q_e = A_m \cdot \tilde{p}_e^{-\sigma} \tag{61}$$

where $\tilde{p}_e$ represents the price paid by consumers. Under the iceberg trade cost assumption, the price paid by consumers is equal to the price received by producers. Here, instead, the price received by producers is $p_{e,m} = \frac{\tilde{p}_{e,m}}{\tau_m}$. Each exporting firm's variable cost is:

$$\frac{1}{\varphi_e} \cdot \frac{1}{\lambda+1} \cdot \left(q_{e,CHN} + \cdot q_{e,ROW}\right)^{\lambda+1} \tag{62}$$

The difference with the baseline model in the main text is that we are not multiplying the quantities in the expression for the variable cost by the iceberg–type tariffs. Total profits in both markets are:

$$\pi_e = p_{e,CHN} \cdot q_{e,CHN} + p_{e,ROW} \cdot q_{e,ROW} - \frac{1}{\varphi_e} \cdot \frac{1}{\lambda+1} \left(q_{e,CHN} + q_{e,ROW}\right)^{\lambda+1} - f_{CHN} - f_{ROW}\,, \tag{63}$$

The export revenue of each US exporting firm in China and the rest of the world is:

$$r_{e,CHN} = \left(\frac{\sigma-1}{\sigma}\right)^{\frac{\sigma-1}{\lambda\sigma+1}} \cdot A_{CHN} \cdot \varphi_e^{\frac{\sigma-1}{\lambda\sigma+1}} \cdot \tau_{CHN}^{\frac{-\sigma(\lambda+1)}{\lambda\sigma+1}} \cdot \left(\frac{\tau_{ROW}^{\sigma}}{A_{CHN} \cdot \tau_{ROW}^{\sigma} + A_{ROW} \cdot \tau_{CHN}^{\sigma}}\right)^{\frac{\lambda(\sigma-1)}{\lambda\sigma+1}} \tag{64}$$

and

$$r_{e,ROW} = \left(\frac{\sigma-1}{\sigma}\right)^{\frac{\sigma-1}{\lambda\sigma+1}} \cdot A_{ROW} \cdot \varphi_e^{\frac{\sigma-1}{\lambda\sigma+1}} \cdot \tau_{ROW}^{\frac{-\sigma(\lambda+1)}{\lambda\sigma+1}} \cdot \left(\frac{\tau_{CHN}^{\sigma}}{A_{ROW} \cdot \tau_{CHN}^{\sigma} + A_{CHN} \cdot \tau_{ROW}^{\sigma}}\right)^{\frac{\lambda(\sigma-1)}{\lambda\sigma+1}} \tag{65}$$

respectively. This framework leads to the same propositions as in the baseline model.

Regarding the response of exports to China to changes in China's tariff, we can use the

expression for an exporter's revenue in the Chinese market in equation (64) to write:

$$\frac{\mathrm{d}r_{e,CHN}}{\mathrm{d}\tau_{CHN}} = \frac{r_{e,CHN}}{\tau_{CHN}} \cdot \frac{\sigma}{\lambda\sigma+1}\left(-(\lambda+1)-\lambda(\sigma-1)\frac{A_{ROW}\tau_{CHN}^{\sigma}}{A_{CHN}\tau_{ROW}^{\sigma}+A_{ROW}\tau_{CHN}^{\sigma}}\right)<0 \tag{66}$$

Regarding the response of exports to the rest of the world to changes in China's tariff, we can use the expression for an exporter's revenue in the rest of the world market in equation (65) to write:

$$\frac{\mathrm{d}r_{e,ROW}}{\mathrm{d}\tau_{CHN}} = \frac{r_{e,ROW}}{\tau_{CHN}} \cdot \frac{\lambda\sigma(\sigma-1)}{\lambda\sigma+1}\left(1+\frac{A_{ROW}\tau_{CHN}^{\sigma}}{A_{CHN}\tau_{ROW}^{\sigma}+A_{ROW}\tau_{CHN}^{\sigma}}\right)>0 \tag{67}$$

A.1.6 Additional results: Alternative model of financial constraints

As a robustness check, we examine an alternative model of financial constraints. We extend the baseline model (we abstract from firm–to–firm trade) and incorporate the role of financial constraints in a framework in which exporting firms have a working capital requirement and face a borrowing limit. This structure is common in the literature and is used, for example, in Benguria and Taylor [2020] to study the response of trade flows to deleveraging shocks.

Specifically, we assume that exporting firms have a working capital requirement, such that they need to borrow funds to finance a fraction $\delta < 1$ of their variable cost before production takes place. There is a binding borrowing constraint that limits the amount that exporting firms can borrow to meet the working capital requirement, such that $\delta C_e \leq \bar{D}$, where C_e stands for the firms' variable cost and $\bar{D}$ represents the borrowing limit.

The binding borrowing constraint implies that the variable cost will be equal to a constant ($\frac{\bar{D}}{\delta}$). This results in a restriction on the sum of the output in both markets, $q_{CHN} + q_{ROW}$. We show below that exports are more responsive to tariffs when the borrowing limit is higher. As in the main text, we express the results in terms of the leverage ratio of an exporting firm (defined as the amount borrowed over total revenue), which as we show below is increasing in the borrowing limit.

We use the ad-valorem tariff assumption described in detail earlier in Appendix Section A.1.5.

In this context, an exporting firm maximizes profits:

$$\pi_e = p_{e,CHN} \cdot q_{e,CHN} + p_{e,ROW} \cdot q_{e,ROW} - \frac{1}{\varphi_e} \cdot \frac{1}{\lambda+1}\left(q_{e,CHN}+q_{e,ROW}\right)^{\lambda+1} - f_{CHN} - f_{ROW}, \tag{68}$$

subject to the borrowing constraint $\delta C_e \leq \bar{D}$, where C_e stands for the exporter's variable cost. Given that variable cost is:

$$\frac{1}{\varphi_e} \cdot \frac{1}{\lambda+1} \cdot \left(q_{e,CHN} + q_{e,ROW}\right)^{\lambda+1} \tag{69}$$

the constraint can be written as:

$$q_{e,CHN} + q_{e,ROW} \leq \left(\frac{\bar{D} \cdot (\lambda+1) \cdot \varphi_e}{\delta}\right)^{\frac{1}{\lambda+1}} \tag{70}$$

Solving this problem yields exports to China:

$$r_{CHN} = A_{CHN} \cdot \tau_{CHN}^{-1} \cdot \left(\frac{\tau_{ROW}^{\sigma}}{A_{ROW}\tau_{CHN}^{\sigma} + A_{CHN}\tau_{ROW}^{\sigma}}\right)^{\frac{\sigma-1}{\sigma}} \cdot \left(\frac{\bar{D} \cdot (\lambda+1) \cdot \varphi_e}{\delta}\right)^{\frac{\sigma-1}{\sigma(\lambda+1)}} \tag{71}$$

and exports to the rest of the world:

$$r_{ROW} = A_{ROW} \cdot \tau_{ROW}^{-1} \cdot \left(\frac{\tau_{CHN}^{\sigma}}{A_{ROW}\tau_{CHN}^{\sigma} + A_{CHN}\tau_{ROW}^{\sigma}}\right)^{\frac{\sigma-1}{\sigma}} \cdot \left(\frac{\bar{D} \cdot (\lambda+1) \cdot \varphi_e}{\delta}\right)^{\frac{\sigma-1}{\sigma(\lambda+1)}} \tag{72}$$

Regarding the response of exports to China to changes in China's tariff:

$$\frac{dr_{e,CHN}}{d\tau_{CHN}} = \frac{r_{e,CHN}}{\tau_{CHN}} \cdot \left(-1 - (\sigma-1) \cdot \frac{A_{ROW}\tau_{CHN}^{\sigma}}{A_{ROW}\tau_{CHN}^{\sigma} + A_{CHN}\tau_{ROW}^{\sigma}}\right) < 0 \tag{73}$$

Regarding the response of exports to the rest of the world to changes in China's tariff:

$$\frac{dr_{e,ROW}}{d\tau_{CHN}} = \frac{r_{e,ROW}}{\tau_{CHN}}(\sigma-1)\frac{A_{CHN} \cdot \tau_{ROW}^{\sigma}}{A_{ROW}\tau_{CHN}^{\sigma} + A_{CHN}\tau_{ROW}^{\sigma}} > 0 \tag{74}$$

Regarding the effect of the borrowing limit on the response of exports to China to changes in China's tariff:

$$\frac{d^2 r_{e,CHN}}{d\tau_{CHN} d\bar{D}} = \frac{\sigma-1}{\sigma(\lambda+1)} \cdot \frac{dr_{e,CHN}}{d\tau_{CHN}} \cdot \frac{1}{\bar{D}} < 0 \tag{75}$$

This implies a higher borrowing limit leads to a larger decline in exports to China in response to an increase in China's tariff.

Regarding the effect of the borrowing limit on the response of exports to the rest of the

world to changes in China's tariff:

$$\frac{d^2 r_{e,ROW}}{d\tau_{CHN} d\bar{D}} = \frac{\sigma - 1}{\sigma(\lambda + 1)} \cdot \frac{d r_{e,ROW}}{d\tau_{CHN}} \cdot \frac{1}{\bar{D}} > 0 \tag{76}$$

This implies a higher borrowing limit leads to a larger increase in exports to the rest of the world in response to an increase in China's tariff.

Finally, we can show that the leverage ratio is increasing in the borrowing limit. The leverage ratio L is the ratio between the amount borrowed ($\bar{D}$) and total revenue:

$$\begin{aligned} L &= \frac{\bar{D}}{r_{e,CHN} + r_{e,ROW}} \\ &= \bar{D}^{\frac{\sigma\lambda+1}{\sigma(\lambda+1)}} \cdot \frac{1}{A_{ROW}\tau_{CHN}^{\sigma} + A_{CHN}\tau_{ROW}^{\sigma}} \cdot \left(\frac{A_{CHN}\tau_{ROW}^{\sigma}}{\tau_{CHN}} + \frac{A_{ROW}\tau_{CHN}^{\sigma}}{\tau_{ROW}} \right) \cdot \left(\frac{(\lambda+1)\varphi_e}{\delta} \right)^{-\frac{\sigma-1}{\sigma(\lambda+1)}} \end{aligned} \tag{77}$$

This implies that i) An increase in China's tariff is associated with a decline in exports to China which is increasing in the firm's leverage ratio; and ii) An increase in China's tariff is associated with an increase in exports to the rest of the world, which is increasing in the firm's leverage ratio.

A.2 Appendix to Section 3

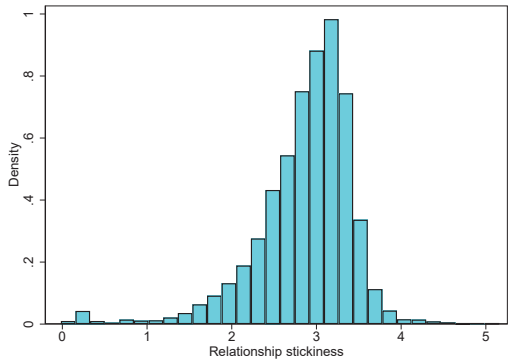
A.2.1 On the measure of relationship stickiness

Here, we describe in more detail the measure of relationship stickiness from Martin et al. [2020]. In addition, we provide a validation exercise that suggests that while this measure is constructed from microdata on French exporters and their foreign importers, it seems applicable to the US context.

- Figure A.1 reports an histogram of the relationship stickiness measure across HS6 product categories (which is the level at which Martin et al. [2020] report this measure and which we use in this paper).

- To assess how applicable is this measure in this context, we examine the duration of exports at the highly disaggregate HS10 product category and show that among products that have an above median degree of relationship stickiness, exports at the HS10 level are much less likely to see breaks over time. Specifically, we consider exports to China between 2015 and 2017 before the trade war. We examine data by month and HS10 product. For each product, we compute the number of months with positive exports. We then divide these products into those with above or below median relationship stickiness according to Martin et al. [2020]'s measure. First, we find that positive exports occur 66% vs. 42% of the months in industries with high vs. low relationship stickiness. Second, we estimate a regression of the number of months with positive exports on relationship stickiness, controlling for the log of exports in each product and by HS2 fixed effects. The results in Table A.1 indicate that a one standard deviation increase in relationship stickiness is associated with a 0.13 standard deviations increase in the number of months with positive exports. In other words, the mean duration of trade spells at the HS10 level is higher in industries with higher relationship stickiness.

Figure A.1: Histogram of the relationship stickiness measure



Notes: This figure reports an histogram of the relationship stickiness measure of Martin et al. [2020] across HS6–digit products.

Table A.1: Relationship stickiness and trade duration

	(1)	(2)
Relationship stickiness	0.161***	0.125***
	(0.008)	(0.009)
Observations	6244	6244

Notes: This table reports the results of a regression in which the dependent variable is the number of months with positive exports over 2015–2017 for each HS10 product exported to China. Both columns control for the log of exports in each HS10 product. Column 2 controls for HS2 fixed effects. ***, **, and * indicate statistical significance at the 1%, 5% and 10% level.

A.2.2 Other Variables

Here, we briefly mention the source and definition of supplementary variables used in the analysis.

1. **Inventory ratios:** We measure the inventory intensity of each industry using COMPUSTAT. We first compute the inventory to revenue ratio of each firm, then compute the median within firms over time between 2012 and 2016, and finally compute the median across firms within NAICS 4–digit industries, which are then matched to HS 10–digit codes.

2. **External finance dependence:** We measure the external finance dependence of each industry using COMPUSTAT. We first compute the ratio of capital expenditure minus cash flow from operations over capital expenditure of each firm, then compute the median within firms over time between 2012 and 2016, and finally compute the median across firms within NAICS 4–digit industries, which are then matched to HS 10–digit codes.

3. **Trade credit:** We measure the trade credit intensity of each industry using COMPUSTAT. We first compute the ratio of accounts receivable minus accounts payable over total assets for each firm, then compute the median within firms over time between 2012 and 2016, and finally compute the median across firms within NAICS 4–digit industries, which are then matched to HS 10–digit codes.

4. **Upstreamness:** Antràs et al. [2012] define and compute a measure of upstreamness that captures the relative position of different industries in supply chains using input-output tables. The data is originally reported using the industry classification of the 2002 input–output table of the US, and assigned to HS10 products using a concordance provided by the Census.

5. **Contract intensity:** Nunn [2007] defines contract intensity as the fraction of a final good's inputs that are relationship–specific. This is constructed by Nunn using input–output tables as the share of value of a final good's inputs that are not sold in organized exchanges and are not referenced priced.

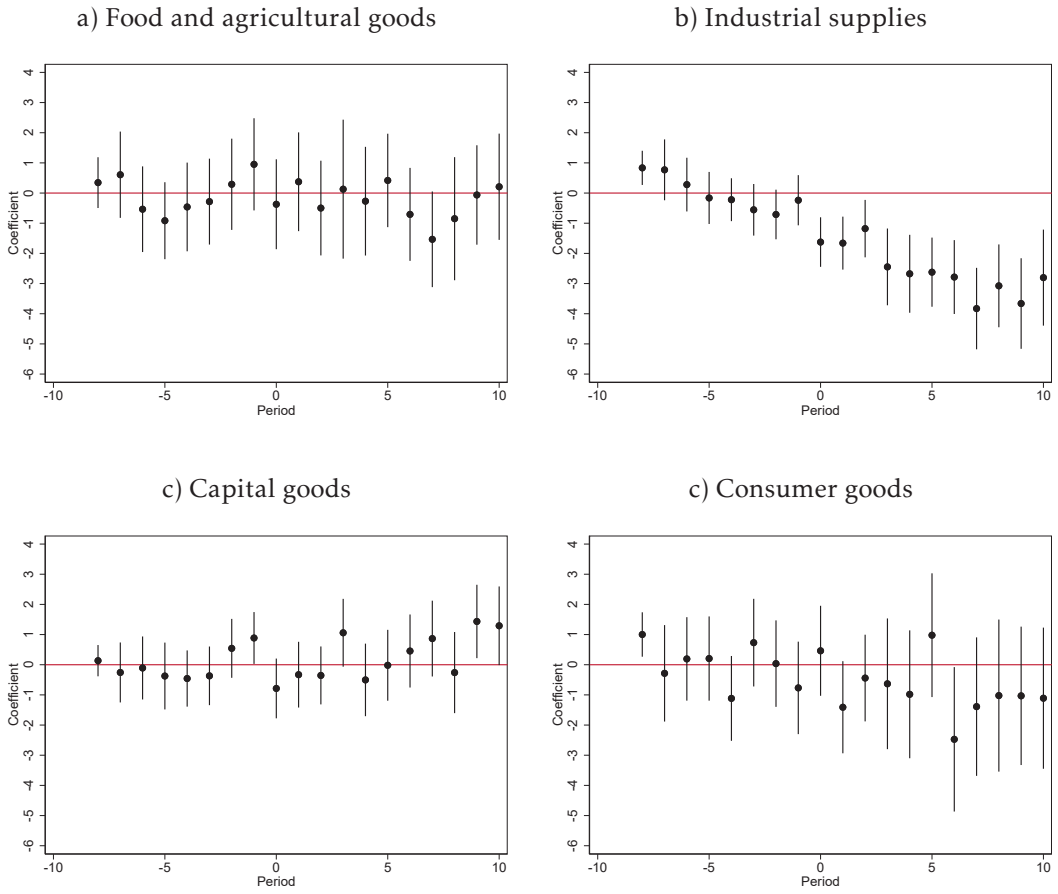
6. **Quality ladders:** Khandelwal [2010] estimates the quality of traded goods at the HS10 level using data on trade volumes and prices and the intuition that conditional on prices,

higher market shares imply higher quality. Khandelwal [2010] defines the length of quality ladders as the range of qualities for each product.

7. **Related-party trade** We compute the share of related-party trade in US exports in 2017 using data reported by the US Census. The data is originally reported at the level of NAICS 6–digit industries, and we use a concordance provided by the US Census Bureau to assign it to HS10 products.

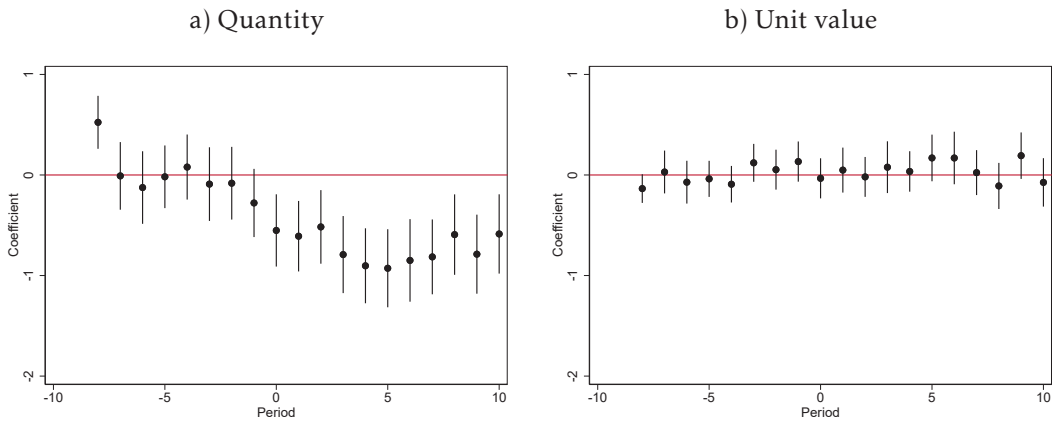
A.3 Appendix to Section 4

Figure A.2: China's retaliatory tariffs and US exports by sector



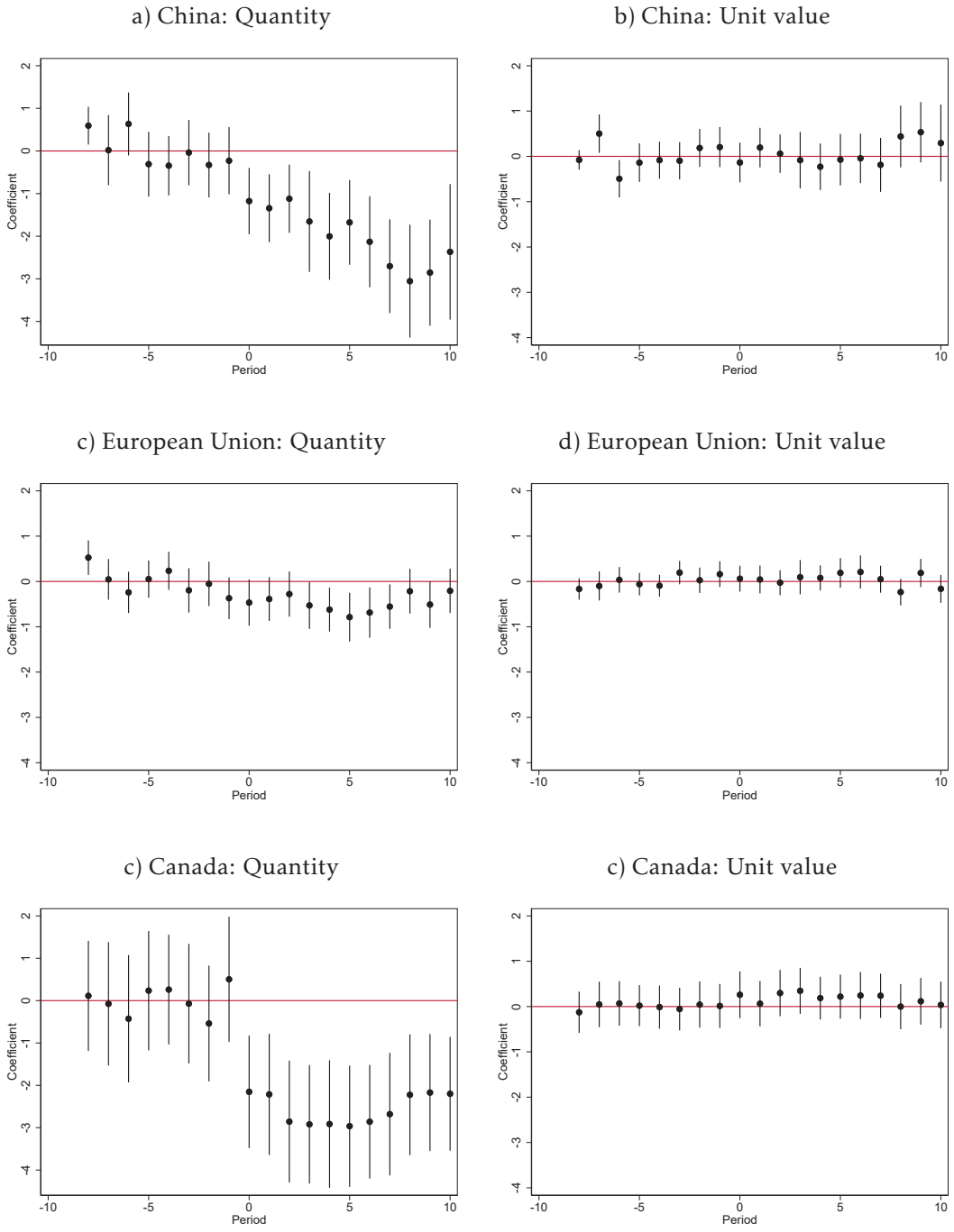
Notes: This figure plots the coefficients obtained from the estimation of equation (9). Vertical bars represent 90% confidence intervals.

Figure A.3: Retaliatory tariffs and US exports: Quantities and unit values



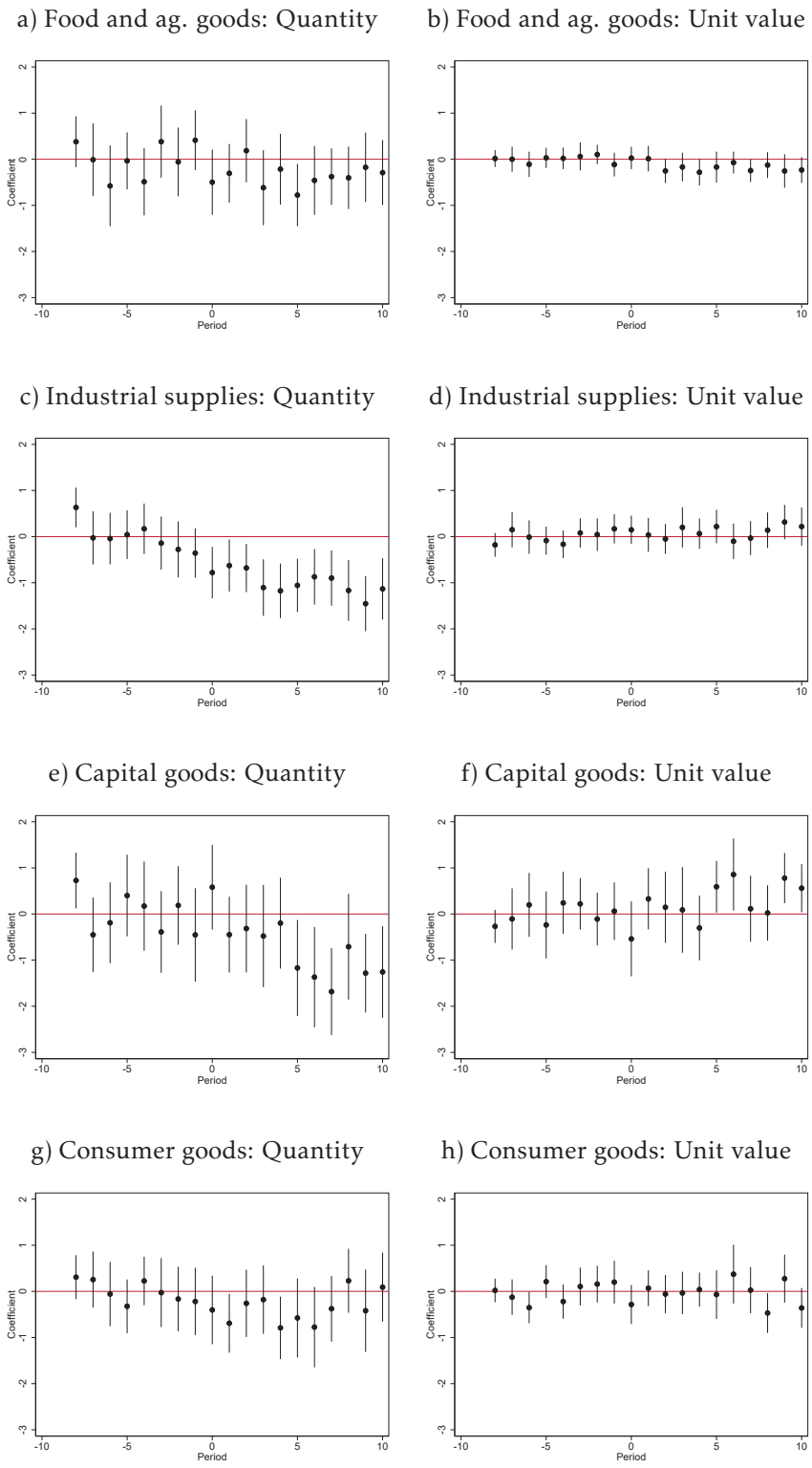
Notes: This figure plots the coefficients obtained from the estimation of equation (9). Vertical bars represent 90% confidence intervals.

Figure A.4: Retaliatory tariffs and US exports to main retaliating countries: Quantities and unit values



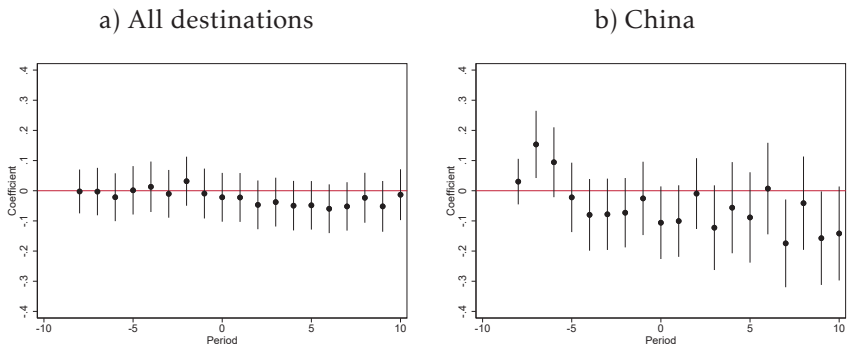
Notes: This figure plots the coefficients obtained from the estimation of equation (9). Vertical bars represent 90% confidence intervals.

Figure A.5: Retaliatory tariffs and US exports by sector: Quantities and unit values



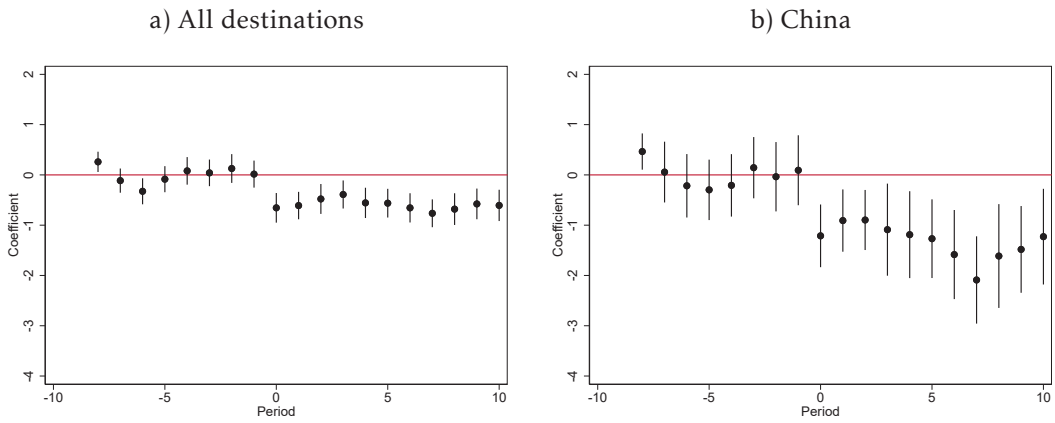
Notes: This figure plots the coefficients obtained from the estimation of equation (9). Vertical bars represent 90% confidence intervals.

Figure A.6: China's retaliatory tariffs and US exports: Extensive margin



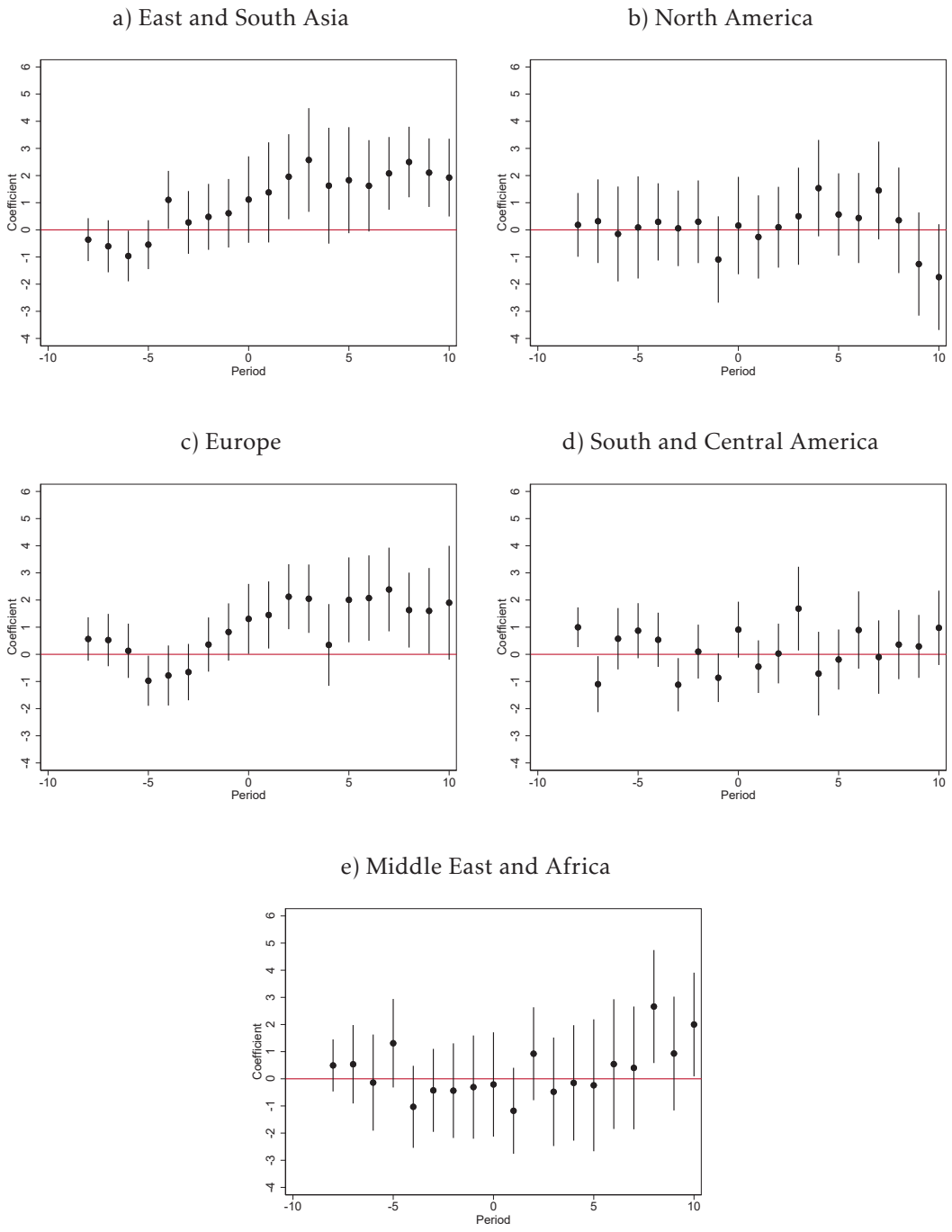
Notes: This figure plots the coefficients obtained from the estimation of equation (9). Vertical bars represent 90% confidence intervals.

Figure A.7: Retaliatory tariffs and US exports: Products defined at the HS 6–digit level



Notes: This figure plots the coefficients obtained from the estimation of equation (9). Vertical bars represent 90% confidence intervals.

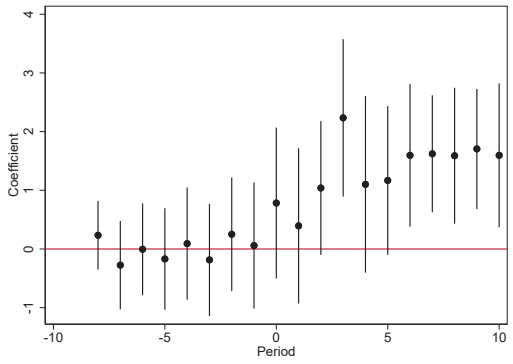
Figure A.8: China's retaliatory tariffs and US exports to ROW: Breakdown by regions



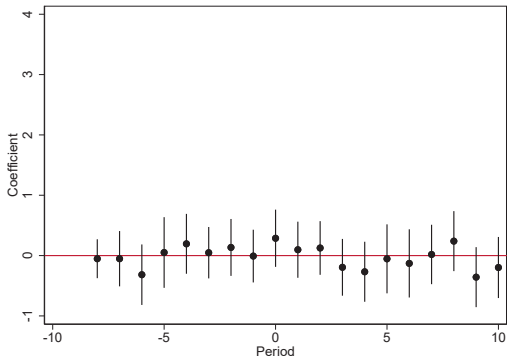
Notes: This figure plots the coefficients obtained from the estimation of equation (10). Vertical bars represent 90% confidence intervals.

Figure A.9: China's retaliatory tariffs US and exports to ROW: Quantities and unit values

a) Chinese tariffs and US exports to ROW: Quantity

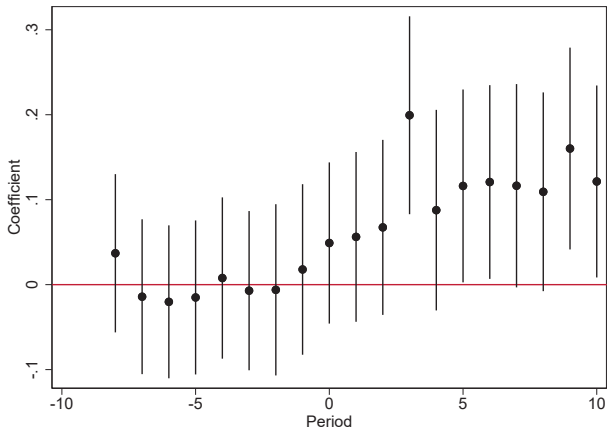


b) Chinese tariffs and US exports to ROW: Unit value



Notes: This figure plots the coefficients obtained from the estimation of equation (10). Vertical bars represent 90% confidence intervals.

Figure A.10: China's Retaliatory Tariffs and US Exports to ROW: Extensive margin



Notes: This figure plots the coefficients obtained from the estimation of equation (10). Vertical bars represent 90% confidence intervals.

A.3.1 Alternative fixed effects

Here, we show that the results in Section 4 are robust to alternative sets of fixed effects. In the case of exports to all destinations (in Figure 2a), the baseline analysis includes country × HS10 product fixed effects and country × HS2 sector × time (year–month) fixed effects. In Appendix Figure A.11 below, we show the results are robust to including the following sets of fixed effects:

- country × HS10 product fixed effects and time (year–quarter) fixed effects.
- country × HS10 product fixed effects and country × time (year–quarter) fixed effects.
- country × HS10 product fixed effects and country × time (year–quarter) fixed effects and HS2 sector × time (year–quarter).
- country × HS10 product fixed effects and country × HS4 industry × time (year–month) fixed effects.
- country × HS10 product fixed effects and country × time (year–quarter) fixed effects and HS10 product × time (year–quarter) fixed effects.

In the case of exports to a China (in Figure 2b), the baseline analysis includes HS10 product fixed effects and HS2 sector × time (year–month) fixed effects. In Appendix Figure A.12 below, we show the results are robust to including the following sets of fixed effects:

- HS10 product fixed effects and time (year–quarter) fixed effects.
- HS10 product fixed effects and HS4 industry × time (year–month) fixed effects.

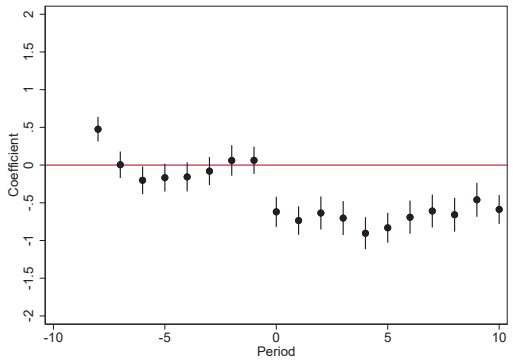
In the case of export reallocation in response to Chinese retaliatory tariffs (in Figure 4), the baseline analysis includes country × HS10 product fixed effects and country × HS2 sector × time (year–month) fixed effects. In Appendix Figure A.13 below, we show the results are robust to including the following sets of fixed effects:

- country × HS10 product fixed effects and time (year–quarter) fixed effects.
- country × HS10 product fixed effects and country × time (year–quarter) fixed effects.

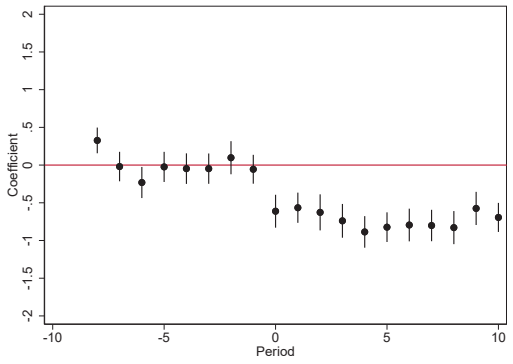
- country × HS10 product fixed effects and country × time (year–quarter) fixed effects and HS2 sector × time (year–quarter).
- country × HS10 product fixed effects and country × HS4 industry × time (year–month) fixed effects.

Figure A.11: Retaliatory tariffs and US exports to all destinations: Alternative fixed effects

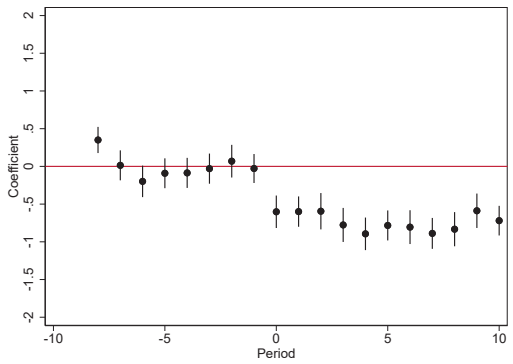
a) Country × HS10 product fixed effects and time (year–quarter) fixed effects



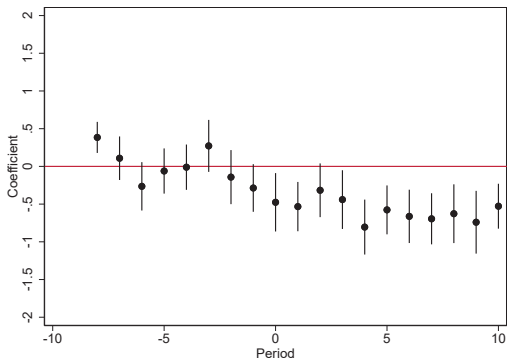
b) Country × HS10 product fixed effects and country × time (year–quarter) fixed effects



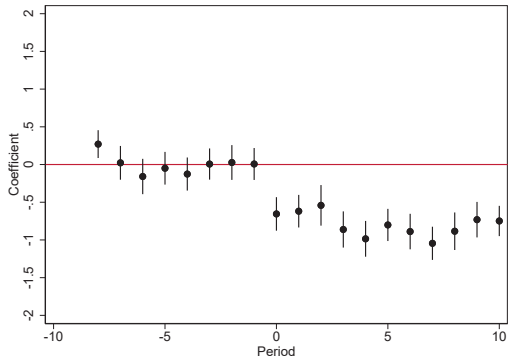
c) Country × HS10 product fixed effects and country × time (year–quarter) fixed effects and HS2 sector × time (year–quarter)



d) Country × HS10 product fixed effects and country × HS4 industry × time (year–month) fixed effects



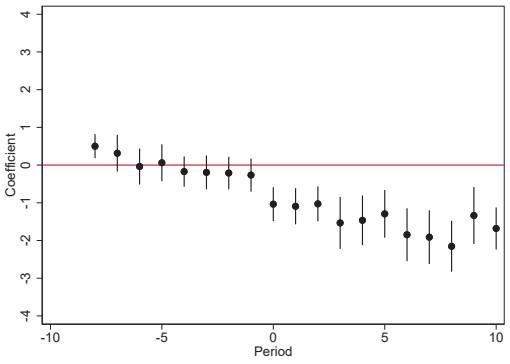
e) Country × HS10 product fixed effects and country × time (year–quarter) fixed effects and HS10 product × time (year–quarter) fixed effects



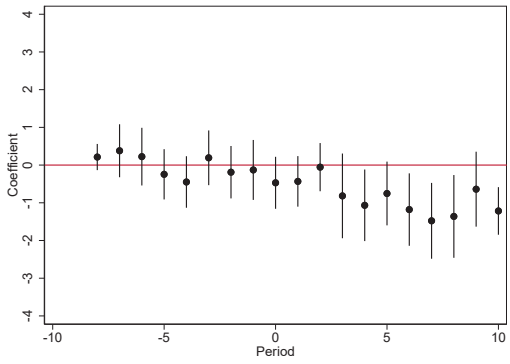
Notes: This figure plots the coefficients obtained from the estimation of equation (9). Vertical bars represent 90% confidence intervals.

Figure A.12: Retaliatory tariffs and US exports to China: Alternative fixed effects

a) HS10 product fixed effects and time (year–quarter) fixed effects



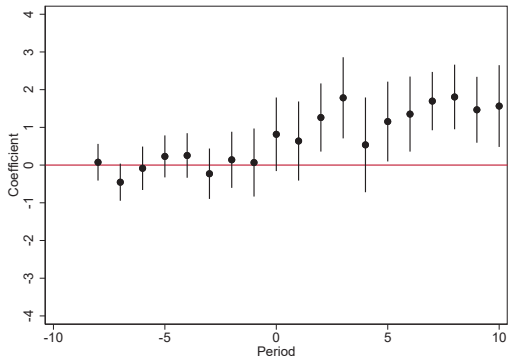
b) HS10 product fixed effects and HS4 industry × time (year–month) fixed effects



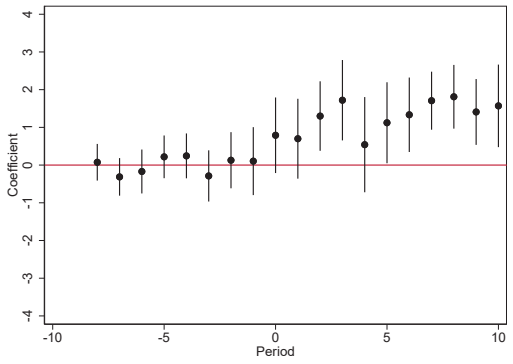
Notes: This figure plots the coefficients obtained from the estimation of equation (9). Vertical bars represent 90% confidence intervals.

Figure A.13: China's retaliatory tariffs and US exports to ROW: Alternative fixed effects

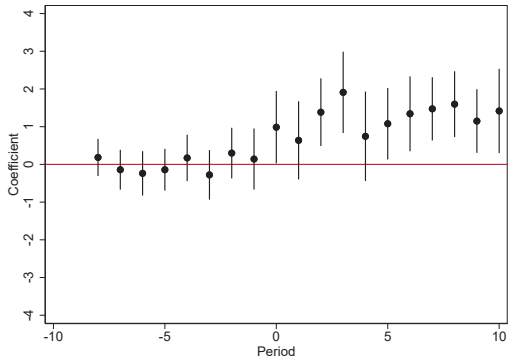
a) Country × HS10 product fixed effects and time (year–quarter) fixed effects



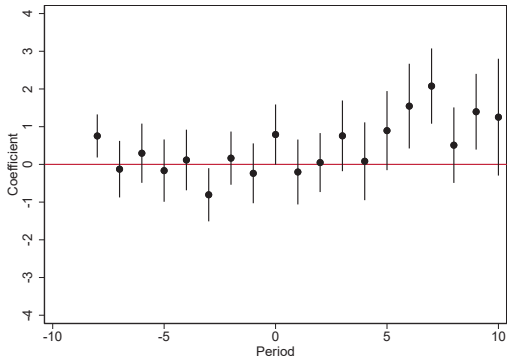
b) Country × HS10 product fixed effects and country × time (year–quarter) fixed effects



c) Country × HS10 product fixed effects and country × time (year–quarter) fixed effects and HS2 sector × time (year–quarter)



d) Country × HS10 product fixed effects and country × HS4 industry × time (year–month) fixed effects

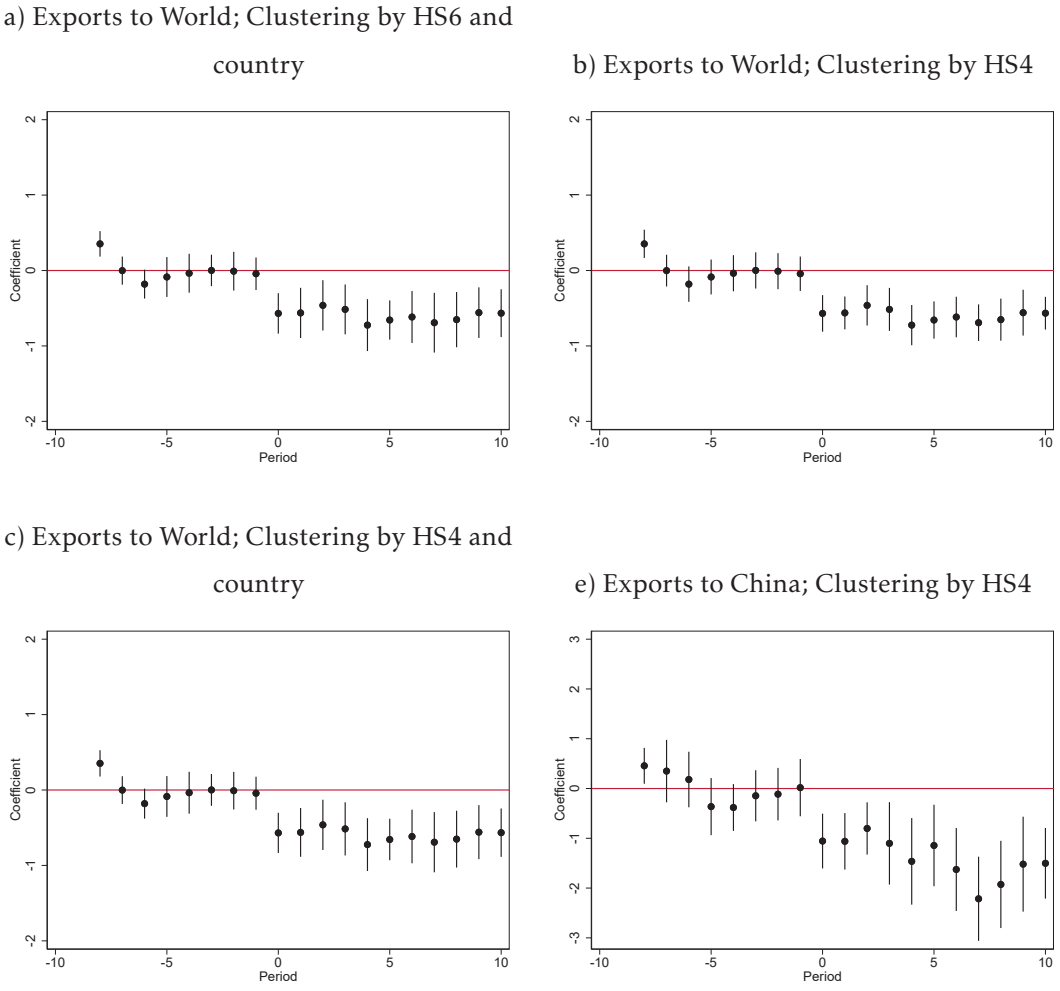


Notes: This figure plots the coefficients obtained from the estimation of equation (10). Vertical bars represent 90% confidence intervals.

A.3.2 Alternative clustering

Here, we show that the results in Section 4 are robust to alternative ways of clustering standard errors. In the baseline analysis in the main text, standard errors are clustered by HS6 product. In Appendix Figures A.14 and A.15 below we cluster standard errors by i) HS4 product, ii) HS6 product and country, using multiway clustering or iiii) HS4 product and country, using multiway clustering. Figure A.14 corresponds to the direct effect of retaliatory tariffs and is equivalent to Figures 2a and 2b in the main text. Figure A.15 corresponds to export reallocation in response to Chinese retaliatory tariffs and is equivalent to Figure 4a in the main text. The results are statistically significant in all cases.

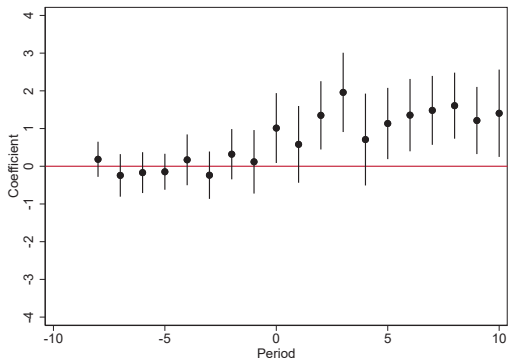
Figure A.14: Retaliatory tariffs and US exports: Alternative clustering



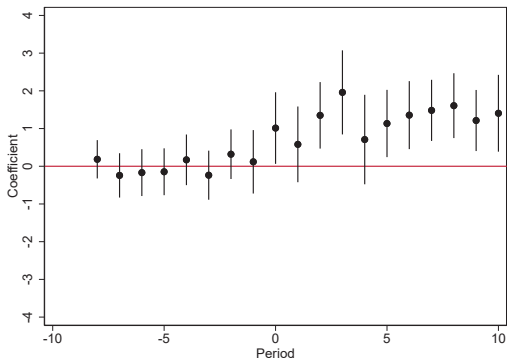
Notes: This figure plots the coefficients obtained from the estimation of equation (9). Vertical bars represent 90% confidence intervals.

Figure A.15: China's retaliatory tariffs and US exports to ROW: Alternative clustering

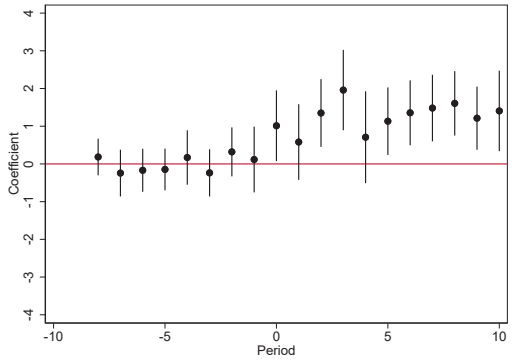
a) Clustering by HS6 and country



b) Clustering by HS4



c) Clustering by HS4 and country



Notes: This figure plots the coefficients obtained from the estimation of equation (10). Vertical bars represent 90% confidence intervals.

A.3.3 Export reallocation in response to other retaliatory tariffs

In Section 4.1 (as well as in Section 5), we document export reallocation toward the rest of the world in response to Chinese tariffs. Here, we extend the same approach and study export reallocation in response to other retaliatory tariffs. We focus on the retaliatory tariffs imposed by Canada, the European Union, and Mexico.[40]

We extend equation (10) to include terms capturing the response of exports to the rest of the world to retaliatory tariffs imposed by Canada, the European Union, and Mexico.

$$\begin{aligned}\log Y_{cpt} = &\sum_{k=--\underline{T}}^{\overline{T}} \beta_k \left(I_{cpk} \times \ln\left(\frac{1+\tau_{cpk}}{1+\tau_{cp0}}\right)\right) + \sum_{k=--\underline{T}}^{\overline{T}} \gamma_k^{CHN} \left(I_{pk}^{CHN} \times \ln\left(\frac{1+\tau_{pk}^{CHN}}{1+\tau_{p0}^{CHN}}\right) \times S_{p0}^{CHN}\right) \\ &+ \sum_{k=--\underline{T}}^{\overline{T}} \gamma_k^{CAN} \left(I_{pk}^{CAN} \times \ln\left(\frac{1+\tau_{pk}^{CAN}}{1+\tau_{p0}^{CAN}}\right) \times S_{p0}^{CAN}\right) + \sum_{k=--\underline{T}}^{\overline{T}} \gamma_k^{EU} \left(I_{pk}^{EU} \times \ln\left(\frac{1+\tau_{pk}^{EU}}{1+\tau_{p0}^{EU}}\right) \times S_{p0}^{EU}\right) \\ &+ \sum_{k=--\underline{T}}^{\overline{T}} \gamma_k^{MEX} \left(I_{pk}^{MEX} \times \ln\left(\frac{1+\tau_{pk}^{MEX}}{1+\tau_{p0}^{MEX}}\right) \times S_{p0}^{MEX}\right) \eta_{cp} + \delta_{cst} + \epsilon_{cpt}\,, \end{aligned} \tag{78}$$

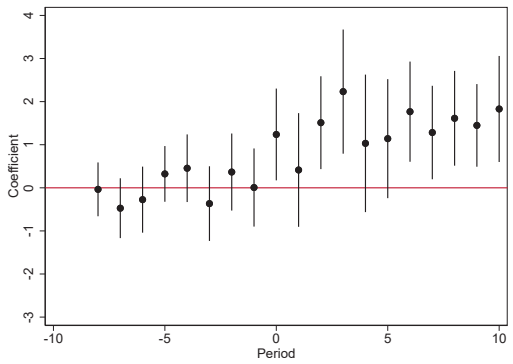
In one approach, we restrict the sample to all export destinations excluding China, Canada, the European Union, and Mexico. The results are shown in Figure A.16. First, we see that the retaliatory tariffs of Canada, the European Union, and Mexico did not lead to a significant amount of export reallocation. This is perhaps to be expected given the relatively small amount of products targeted by these tariffs. Second, the figure shows that the export reallocation effect in response to Chinese tariffs that we document in the main text persists under this augmented approach that also considers other retaliatory tariffs.

In a second approach, we include all export destinations. The results are shown in Figure A.17. They confirm the previous message, with little signs reallocation in response to Canadian, European and Mexican tariffs and substantial reallocation in response to Chinese tariffs.

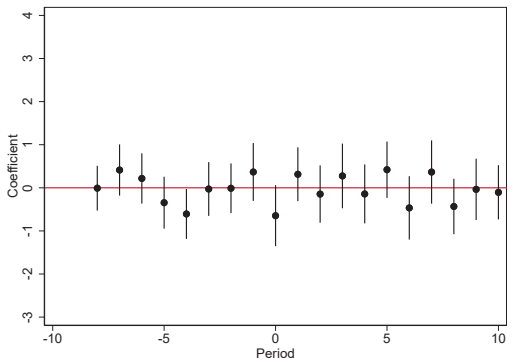
[40] Retaliatory tariffs imposed by other countries were very minor relative to these.

Figure A.16: Retaliatory tariffs and US export reallocation

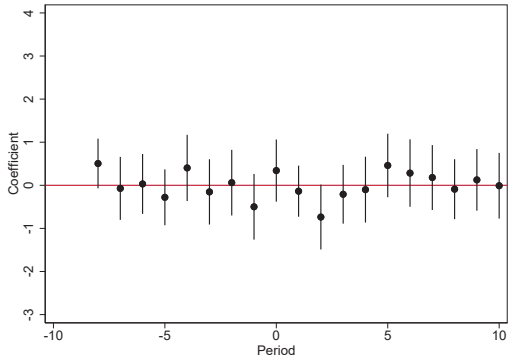
a) Reallocation in response to China's tariffs



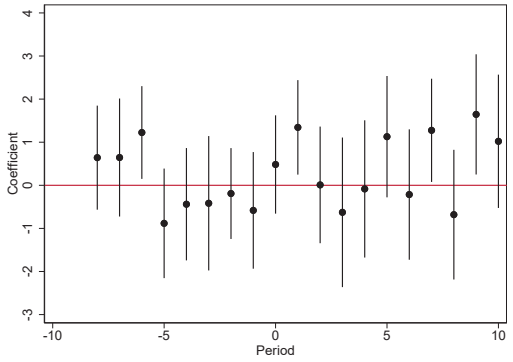
b) Reallocation in response to Canada's tariffs



c) Reallocation in response to the European Union's tariffs



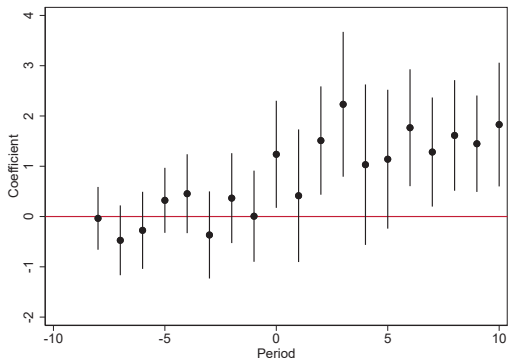
d) Reallocation in response to the Mexico's tariffs



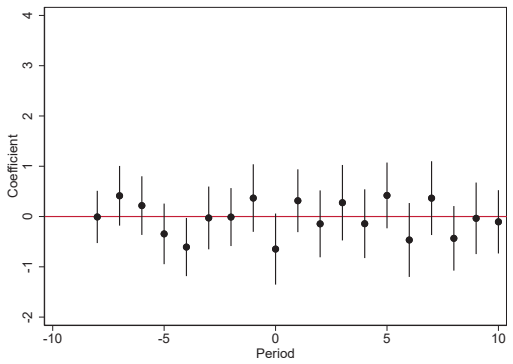
Notes: This figure plots the coefficients obtained from the estimation of equation (78). Vertical bars represent 90% confidence intervals. The sample considers exports to all destinations excluding China, Canada, and the European Union.

Figure A.17: Retaliatory tariffs and US export reallocation

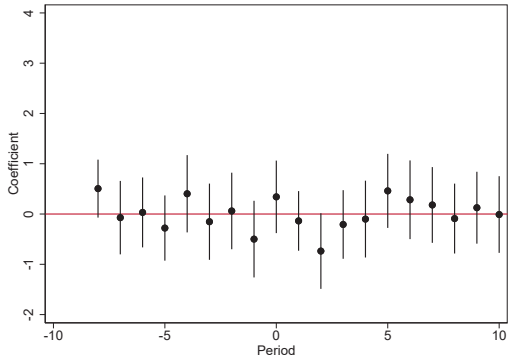
a) Reallocation in response to China's tariffs



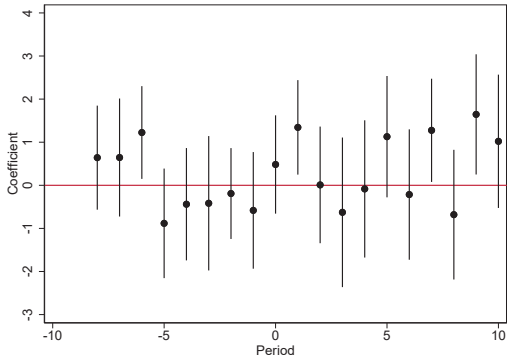
b) Reallocation in response to Canada's tariffs



c) Reallocation in response to the European Union's tariffs



c) Reallocation in response to Mexico's tariffs



Notes: This figure plots the coefficients obtained from the estimation of equation (78). Vertical bars represent 90% confidence intervals. The sample considers all destinations.

A.3.4 Using the De Chaisemartin and d'Haultfoeuille [2020] estimator

We estimate regressions with two–way fixed effects (for products and time or destination–product and time). De Chaisemartin and d'Haultfoeuille [2020] show that in the two period and two group case, these regressions match the difference in difference interpretation and the assumption required to obtain an unbiased estimate of the average treatment effect is the parallel trends assumptions. De Chaisemartin and d'Haultfoeuille [2020] establish that in more general settings including multiple periods, variation in treatment timing, and non-binary treatment (among other features) an additional assumption is required. This second assumption is that the treatment effect is constant across groups and over time. Because this assumption is unlikely to hold, they propose an alternative estimator.

Our setting has these more general features. For this reason, here we report results using the DID_M estimator proposed by De Chaisemartin and d'Haultfoeuille [2020], that is valid when the treatment effect is heterogeneous across groups and/or time. The DID_M estimator estimates the average treatment effect across groups and time based on changes in treatment between periods $t-1$ and t. We focus on exports to China and report the results in Table A.2. In line with the results shown in the main text (Figure 2b), we find a substantial decline in exports to China in response to Chinese tariffs, with a similar magnitude to the effect of the tariff on impact on that figure.

Table A.2: China's Retaliatory Tariffs and US Exports to China: DID_M estimator

	(1)
ATE	-0.788*
	(0.412)
Observations	20949

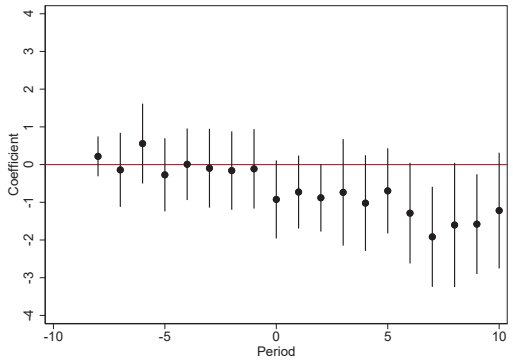
Notes: This table reports the results of using the DID_M estimator of De Chaisemartin and d'Haultfoeuille [2020] for equation (9) on the sample of exports to China including HS10 product and time fixed effects. ***, **, and * indicate statistical significance at the 1%, 5% and 10% level.

A.4 Appendix to Section 5

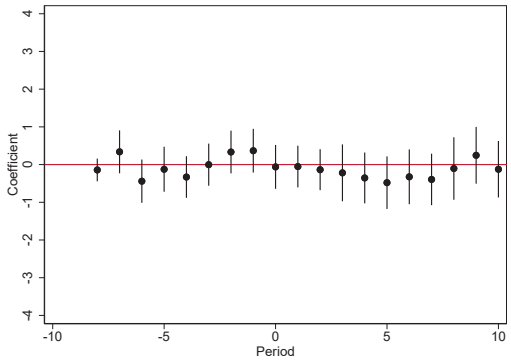
A.4.1 Additional results: Leverage

Figure A.18: China's retaliatory tariffs and US exports to China in high and low leverage industries: Quantities and unit values

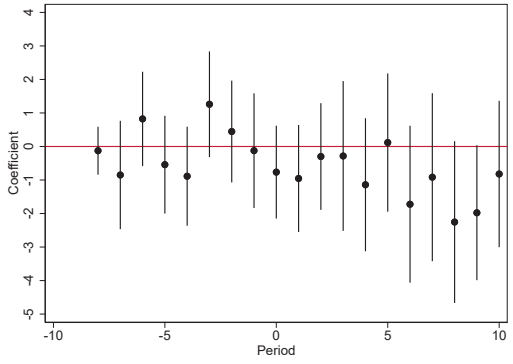
a) High leverage: Quantity



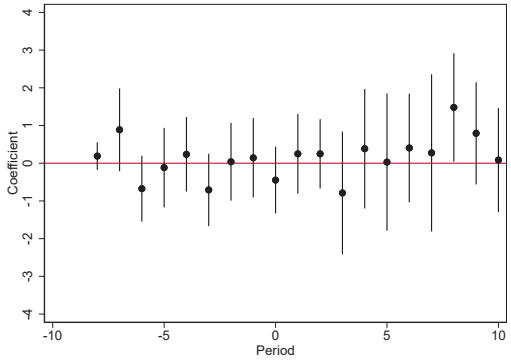
b) High leverage: Unit value



c) Low leverage: Quantity

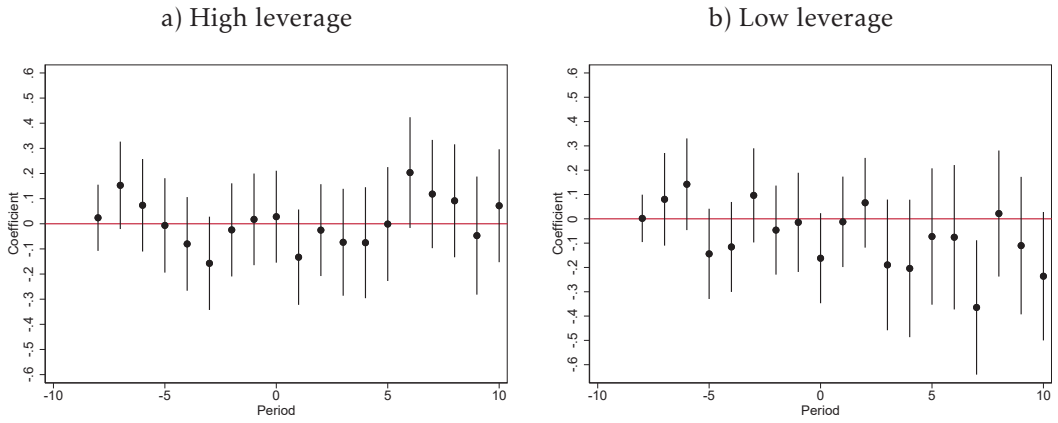


d) Low leverage: Unit value



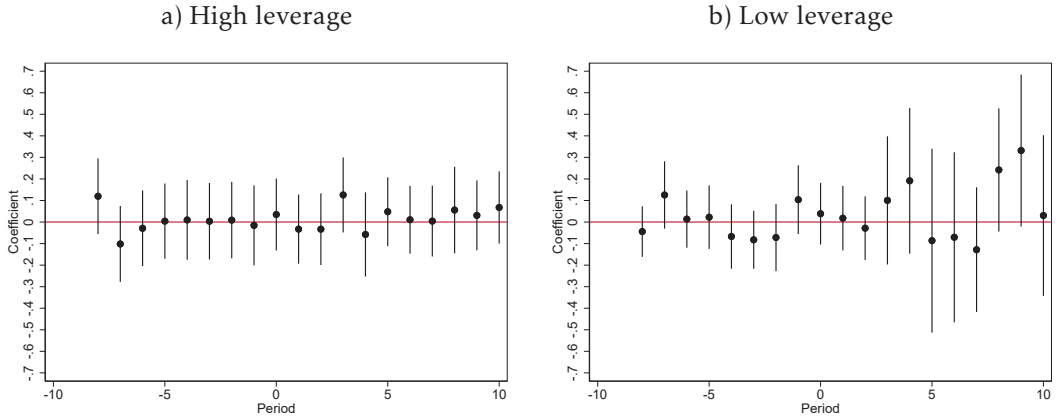
Notes: This figure plots the coefficients obtained from the estimation of equation (9). Vertical bars represent 90% confidence intervals.

Figure A.19: China's retaliatory tariffs and US exports to China in high and low leverage industries: Extensive margin



Notes: This figure plots the coefficients obtained from the estimation of equation (9). Vertical bars represent 90% confidence intervals.

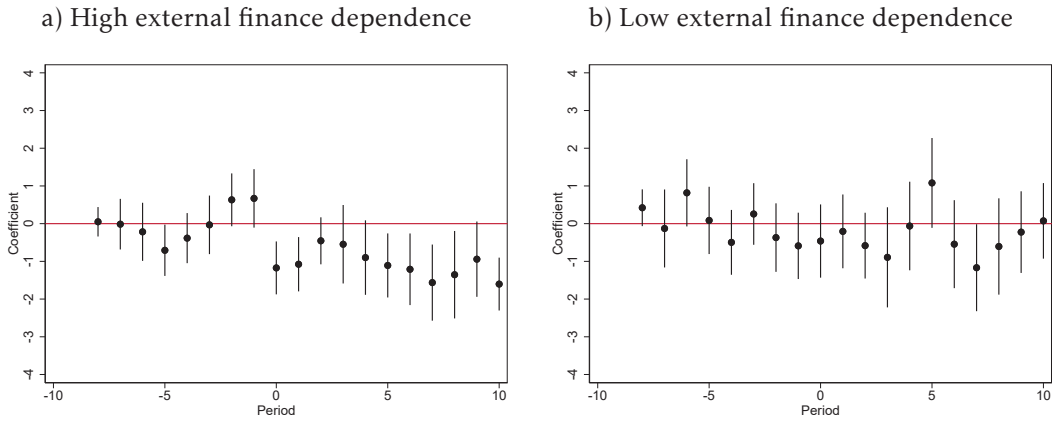
Figure A.20: China's retaliatory tariffs and US exports to ROW in high and low leverage industries: Extensive margin



Notes: This figure plots the coefficients obtained from the estimation of equation (10). Vertical bars represent 90% confidence intervals.

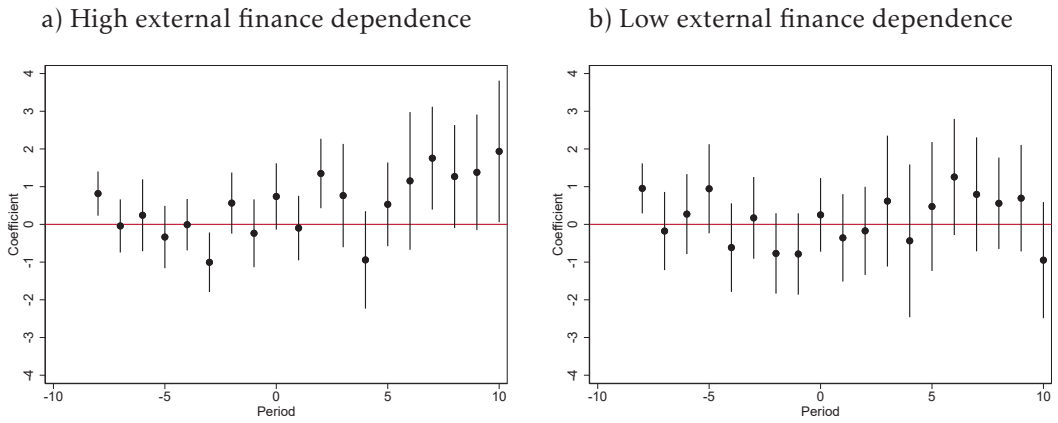
A.4.2 Additional financial measures

Figure A.21: China's retaliatory tariffs and US exports to China: External finance dependence



Notes: This figure plots the coefficients obtained from the estimation of equation (9). Vertical bars represent 90% confidence intervals.

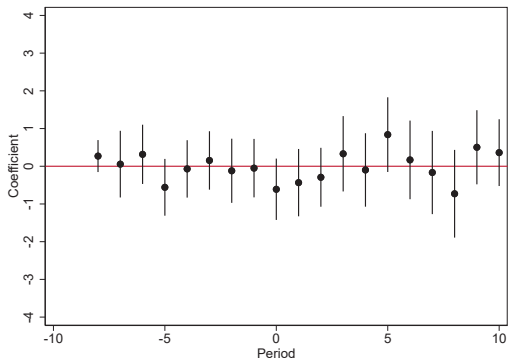
Figure A.22: China's retaliatory tariffs and US exports to ROW: External finance dependence



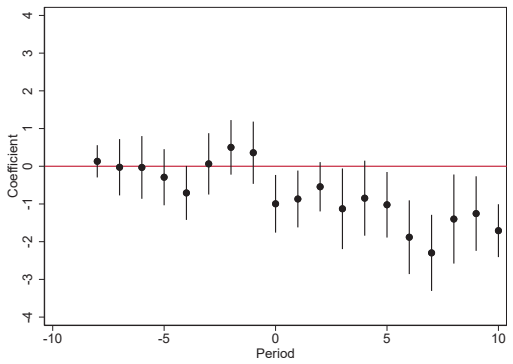
Notes: This figure plots the coefficients obtained from the estimation of equation (10). Vertical bars represent 90% confidence intervals.

Figure A.23: China's retaliatory tariffs and US exports to China: Trade credit

a) High trade credit



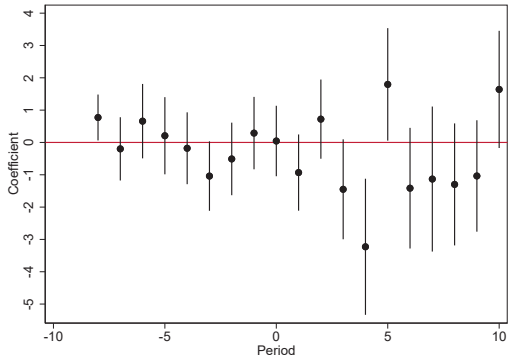
b) Low trade credit



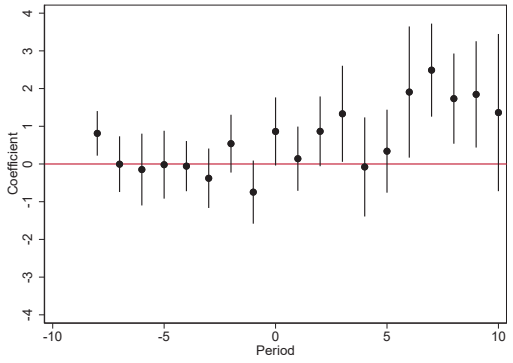
Notes: This figure plots the coefficients obtained from the estimation of equation (9). Vertical bars represent 90% confidence intervals.

Figure A.24: China's retaliatory tariffs and US exports to ROW: Trade credit

a) High trade credit



b) Low trade credit

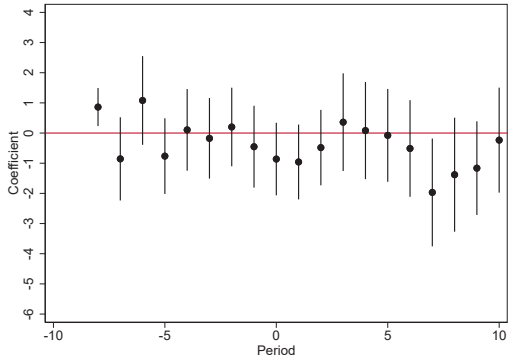


Notes: This figure plots the coefficients obtained from the estimation of equation (10). Vertical bars represent 90% confidence intervals.

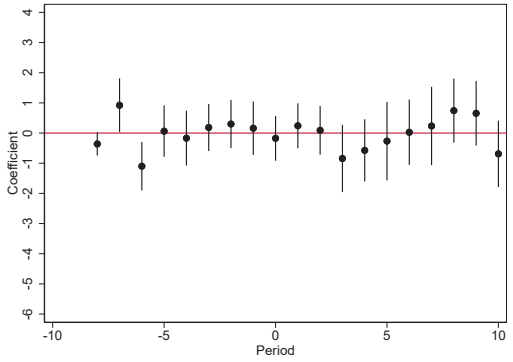
A.4.3 Additional results: Relationship stickiness

Figure A.25: China's retaliatory tariffs and US exports to China in industries with high and low relationship stickiness: Quantities and unit values

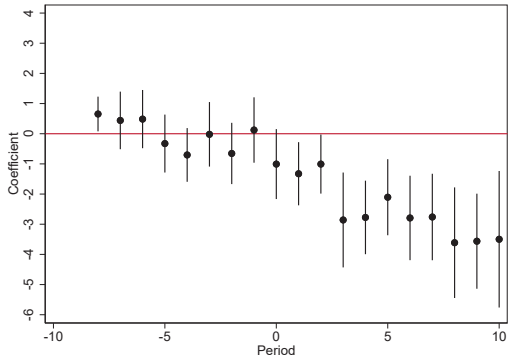
a) High relationship stickiness: Quantity



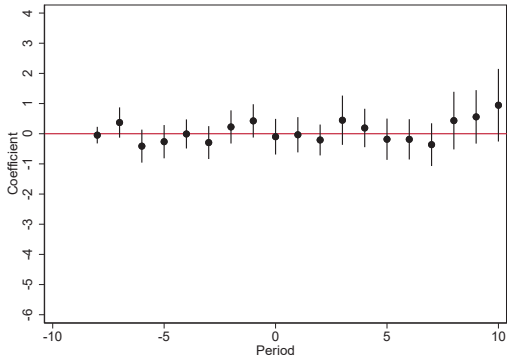
b) High relationship stickiness: Unit value



c) Low relationship stickiness: Quantity



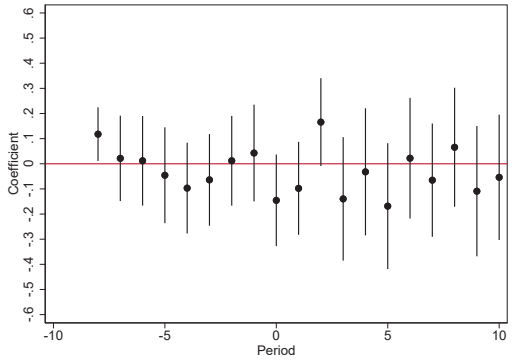
d) Low relationship stickiness: Unit value



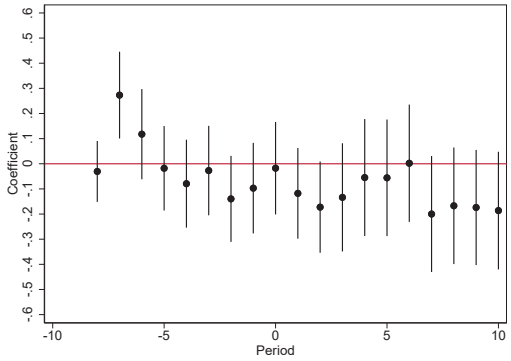
Notes: This figure plots the coefficients obtained from the estimation of equation (9). Vertical bars represent 90% confidence intervals.

Figure A.26: China's retaliatory tariffs and US exports to China in industries with high and low relationship stickiness: Extensive margin

a) High relationship stickiness



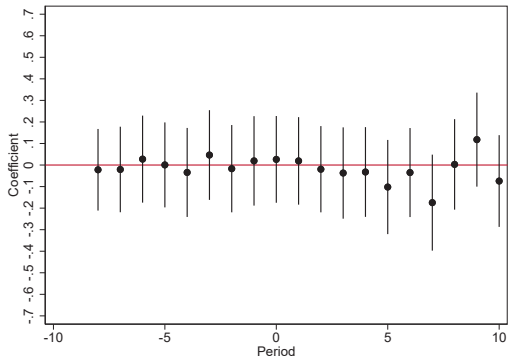
b) Low relationship stickiness



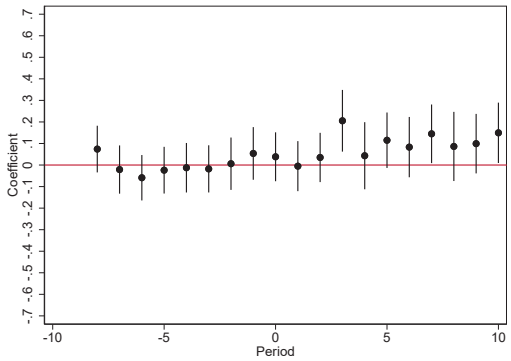
Notes: This figure plots the coefficients obtained from the estimation of equation (9). Vertical bars represent 90% confidence intervals.

Figure A.27: China's retaliatory tariffs and US exports to ROW in industries with high and low relationship stickiness: Extensive margin

a) High relationship stickiness



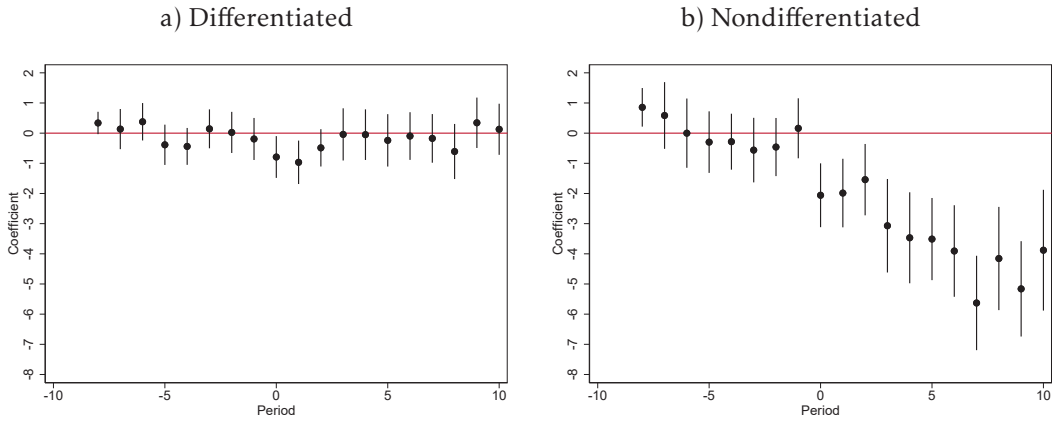
b) Low relationship stickiness



Notes: This figure plots the coefficients obtained from the estimation of equation (10). Vertical bars represent 90% confidence intervals.

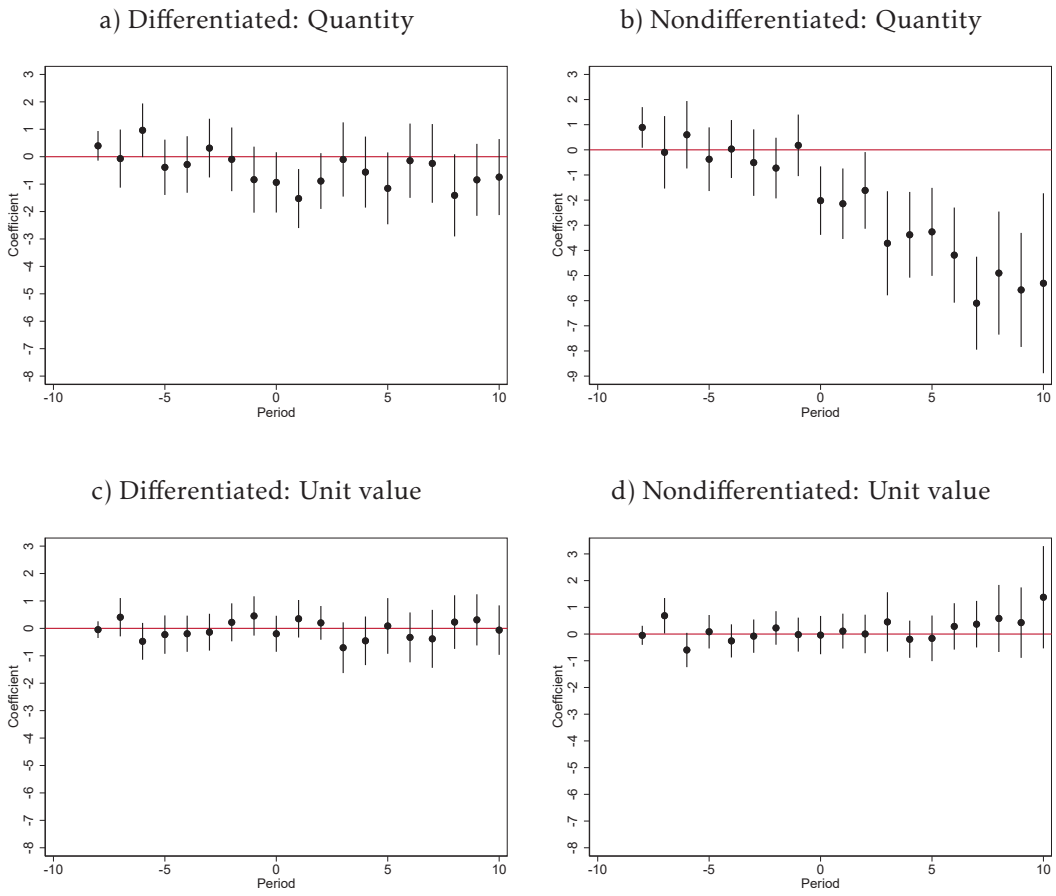
A.4.4 Additional results: Product differentiation

Figure A.28: China's retaliatory tariffs and US exports to China: Product differentiation



Notes: This figure plots the coefficients obtained from the estimation of equation (9). Vertical bars represent 90% confidence intervals.

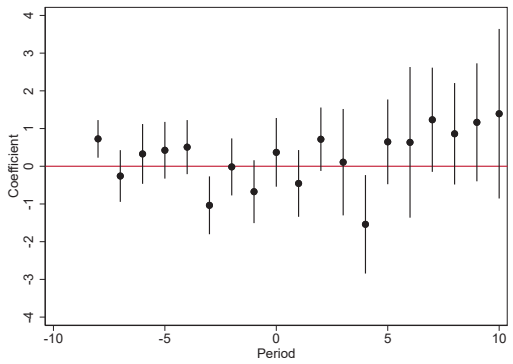
Figure A.29: China's retaliatory tariffs and US exports to China: Product differentiation



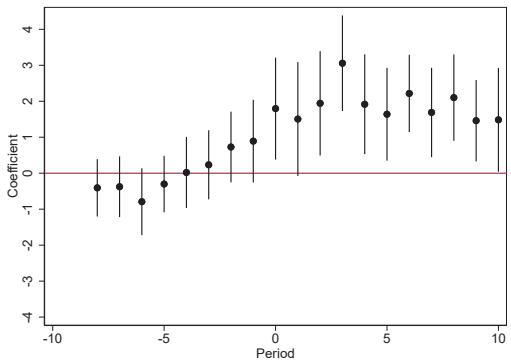
Notes: This figure plots the coefficients obtained from the estimation of equation (9). Vertical bars represent 90% confidence intervals.

Figure A.30: China's retaliatory tariffs and US exports to ROW: Product differentiation

a) Differentiated



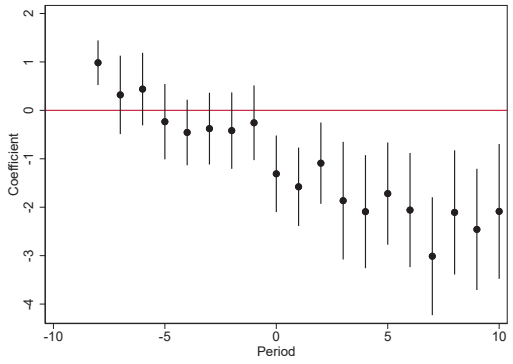
b) Nondifferentiated



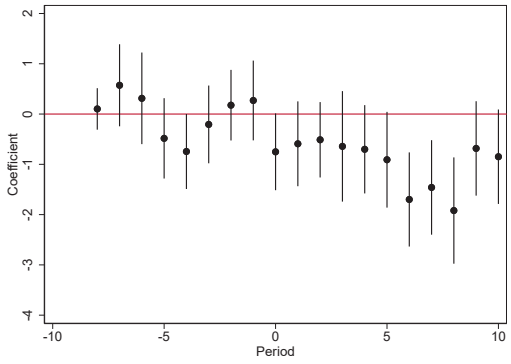
Notes: This figure plots the coefficients obtained from the estimation of equation (9). Vertical bars represent 90% confidence intervals.

Figure A.31: China's retaliatory tariffs and US exports to China: High vs. low elasticities of substitution

a) High elasticity of substitution (lower differentiation)



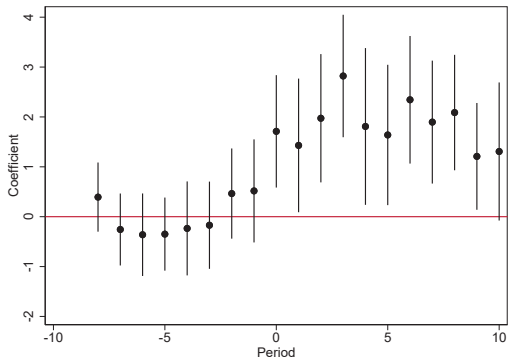
b) Low elasticity of substitution (higher differentiation)



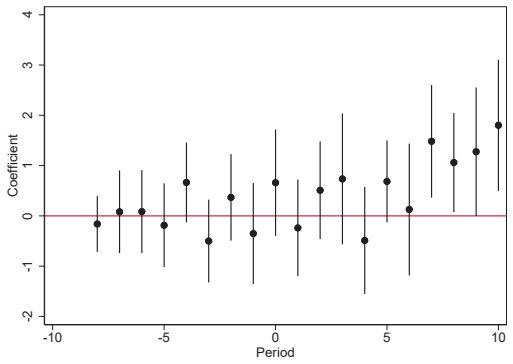
Notes: This figure plots the coefficients obtained from the estimation of equation (9). Vertical bars represent 90% confidence intervals.

Figure A.32: China's retaliatory tariffs and US exports to ROW: High vs. low elasticity of substitution

a) High elasticity of substitution (lower differentiation)



b) Low elasticity of substitution (higher differentiation



Notes: This figure plots the coefficients obtained from the estimation of equation (10). Vertical bars represent 90% confidence intervals.